

From signatures to populations: when does single-cell information improve drug retrieval?

Abstract

Artificial intelligence virtual cells (AIVCs) aim not only to predict how cells respond to perturbations, but ultimately to help prioritize which interventions to test. Transcriptomic drug retrieval has traditionally represented each perturbation by a single expression signature, whereas single-cell perturbation atlases now make it possible to compare entire response populations. Here we formulate population-to-population drug retrieval and ask when this additional single-cell information improves the ranking of candidate drugs. Within the score family studied here, classical mean-signature retrieval is the zero-variance limit of population retrieval, directly connecting the two regimes. Population-level retrieval captures substantial signal beyond the mean, but capturing more information does not necessarily produce a better drug ranking. When the same rankings are judged by criteria less tied to the retrieval score, large gains under response matching disappear or reverse under mechanism-of-action recovery, and most of the advantage that remains against an external functional readout can be reproduced by response magnitude alone. More strikingly, two external protein-based evaluators built from the same cells favour opposite methods when only the statistical form of the evaluator is changed. Together, these results establish population-to-population retrieval as a distinct regime for single-cell drug search and show that its value depends on whether heterogeneous responses are present, recoverable and consequential enough to change candidate choice.

Keywords: artificial intelligence virtual cell; single-cell perturbation; drug retrieval; population-to-population retrieval; cellular heterogeneity; objective–utility mismatch

Introduction

Artificial intelligence virtual cells (AIVCs)^{1,2} are being developed not only to predict how cells respond to perturbations^{3–5}, but ultimately to help decide which perturbations are worth testing. These goals define two complementary problems. Forward prediction asks what response a perturbation will produce^{6–10}; retrieval asks which perturbation best matches a cellular response of interest. Transcriptomic drug retrieval has traditionally addressed the latter by representing each drug with a single perturbation signature and comparing these signatures by similarity or reversal^{11–13}. This representation has made large drug libraries searchable, but it compresses the diversity of responses measured by single-cell experiments into one summary vector. Single-cell perturbation atlases now make it possible to retain that diversity^{14–16}, including differences between cell states¹⁷, minority responses¹⁸ and multiple response patterns within the same population. Drug retrieval can therefore move from comparing mean signatures to comparing entire response populations. We refer to this as population-to-population drug retrieval and ask when this additional single-cell information actually improves the ranking of candidate drugs.

39 More detailed information, however, is not automatically more useful information. Two drugs can
40 produce similar average responses while affecting different subsets of cells, and a population-aware
41 method can distinguish them even when a mean-based method cannot. But this extra discrimination
42 matters only if it leads to a better drug ranking under a biologically relevant criterion. This creates
43 a fundamental evaluation problem^{19–21}. If a distribution-aware method^{22–24} is judged mainly by how
44 well it matches full response distributions^{25,26}, then it is being evaluated on the same type of information
45 it was designed to preserve; a large gain may therefore show that the method captures distributional
46 structure well without showing that it makes a better drug choice^{27,28}. The reverse can also happen if
47 the evaluator reduces the response to a mean, thereby favouring mean-based retrieval²⁹. We refer to this
48 gap between capturing additional information and demonstrating its value for the final ranking as an
49 objective–utility mismatch. The key question is therefore not whether population-level retrieval can use
50 more single-cell information, but when that information is genuinely relevant to the retrieval decision.

51 To answer this question, the information used for retrieval must be separated from the criterion
52 used to judge the result. We therefore formulate drug retrieval directly between single-cell response
53 populations and use a controlled design in which the query, candidate library and ranked list remain
54 unchanged while only the evaluation criterion is varied. Mean-signature retrieval is not a separate task
55 in this framework: within the score family considered here, it is the zero-variance limit of population
56 retrieval, linking classical signature matching to full population comparison. We then judge the same
57 rankings at increasing distance from the retrieval objective, from response-matching criteria, to biolog-
58 ical criteria such as mechanism recovery, and finally to external functional readouts^{30,31}. This design
59 allows us to ask directly whether retaining population structure merely improves how well a response
60 is represented, or whether it improves the ranking of candidate drugs.

61 The two are not equivalent. Population-level retrieval captures substantial signal beyond mean sig-
62 natures, showing that single-cell response populations contain retrievable information that averaging
63 removes. But much of this apparent advantage weakens when the same rankings are judged by criteria
64 that are less tied to the retrieval score: gains under response-matching evaluation disappear or reverse
65 under mechanism recovery, and most of the advantage that remains against an external functional read-
66 out can be reproduced by a simple measure of response magnitude that uses no distributional informa-
67 tion. More strikingly, two external protein-based evaluators built from the same cells favour opposite
68 methods when only the statistical form of the evaluator is changed. External evaluation, in other words,
69 is not automatically neutral. Together, these results suggest a practical framework for when population-
70 level retrieval can add value beyond a mean signature: different cell states must respond differently,
71 those differences must be recoverable from the observed population, and the additional information
72 must ultimately be important enough to change which candidate is preferred. Population-to-population
73 retrieval therefore defines a distinct regime for single-cell drug search, but its value depends not simply
74 on preserving more cellular detail, but on whether that detail changes the decision.

75 Results

76 Population-to-population retrieval generalizes classical signature matching

77 We first asked what changes when the object used for retrieval is a population of single cells rather
78 than a collapsed expression signature. For a query population Q and a library of candidate perturba-

79 tion populations $\{P_d\}$, population-to-population retrieval ranks candidates according to the similarity
80 of their cellular responses (Fig. 1a). Classical mean-signature retrieval^{11,12} performs the same task after
81 replacing each population with a single differential-expression vector. Population-level retrieval instead
82 retains the individual cells and can therefore use response structure that is lost on averaging, including
83 differences between cellular states or subpopulations. The retrieval problem is unchanged; what changes
84 is how much of the measured response is retained before candidate ranking.

85 This distinction also changes what must be shown for the additional information to be useful. Two
86 candidate populations can have the same mean response but differ markedly in how individual cells
87 respond (Fig. 1b–d). A population-aware score can distinguish such candidates, whereas a mean-based
88 score cannot. However, a gain measured by a criterion that rewards the same population structure estab-
89 lishes that the structure has been captured, not necessarily that the resulting ranking is more biologically
90 useful. We therefore use the same candidate rankings throughout and judge them under criteria that are
91 progressively less tied to the retrieval score: response matching, task-proximal biological criteria such as
92 mechanism recovery, and externally measured functional readouts (Fig. 1e,f). This controlled evaluation
93 separates the information used to construct a ranking from the criterion used to judge it.

Mean-signature and population-level retrieval are not separate tasks. Within the score family consid-
ered here, they are connected by the amount of within-population variation retained. Writing a cellular
response as

$$X = \mu + \lambda\epsilon, \quad E[\epsilon] = 0,$$

94 the parameter λ controls the variation around the population mean. As λ approaches zero, the popula-
95 tion collapses onto μ and the corresponding population comparison reduces to its mean-only endpoint
96 (Fig. 1g). Mean-signature retrieval is therefore the zero-variance limit of population retrieval within this
97 framework.

98 This connection is exact for the implemented mean-cosine baseline. Mean-cosine retrieval and the im-
99 plemented CMap-style cosine score apply the same operation to the same mean differential-expression
100 vectors and consequently give identical Hit@1 (0.388492) and identical scores at every query–candidate
101 pair (Spearman $\rho = 1.000$ over 54,180 scores; Fig. 2g). The canonical L1000 weighted connectivity score
102 (WTCS) is related but not identical: it uses rank-based enrichment and yields a different Hit@1 (0.4635).
103 Thus, the equivalence concerns the cosine connectivity rule used here rather than every signature-based
104 retrieval score.

105 The same continuum extends beyond the zero-variance limit. In the subpopulation-coverage score, a
106 temperature parameter β controls how strongly poorly matched subpopulations influence the final dis-
107 tance. On the controlled benchmark, increasing β moves the score continuously from mean-like aggrega-
108 tion (0.7125 as $\beta \rightarrow 0$) towards the worst-matched subpopulation (1.300 as $\beta \rightarrow \infty$; Fig. 1h). The global
109 energy distance is recovered as the single-partition ($K = 1$) case of the same construction. These rela-
110 tionships place classical signature matching and population comparison on a common axis of sensitivity
111 to within-population structure rather than treating them as unrelated method classes (Supplementary
112 Table 2).

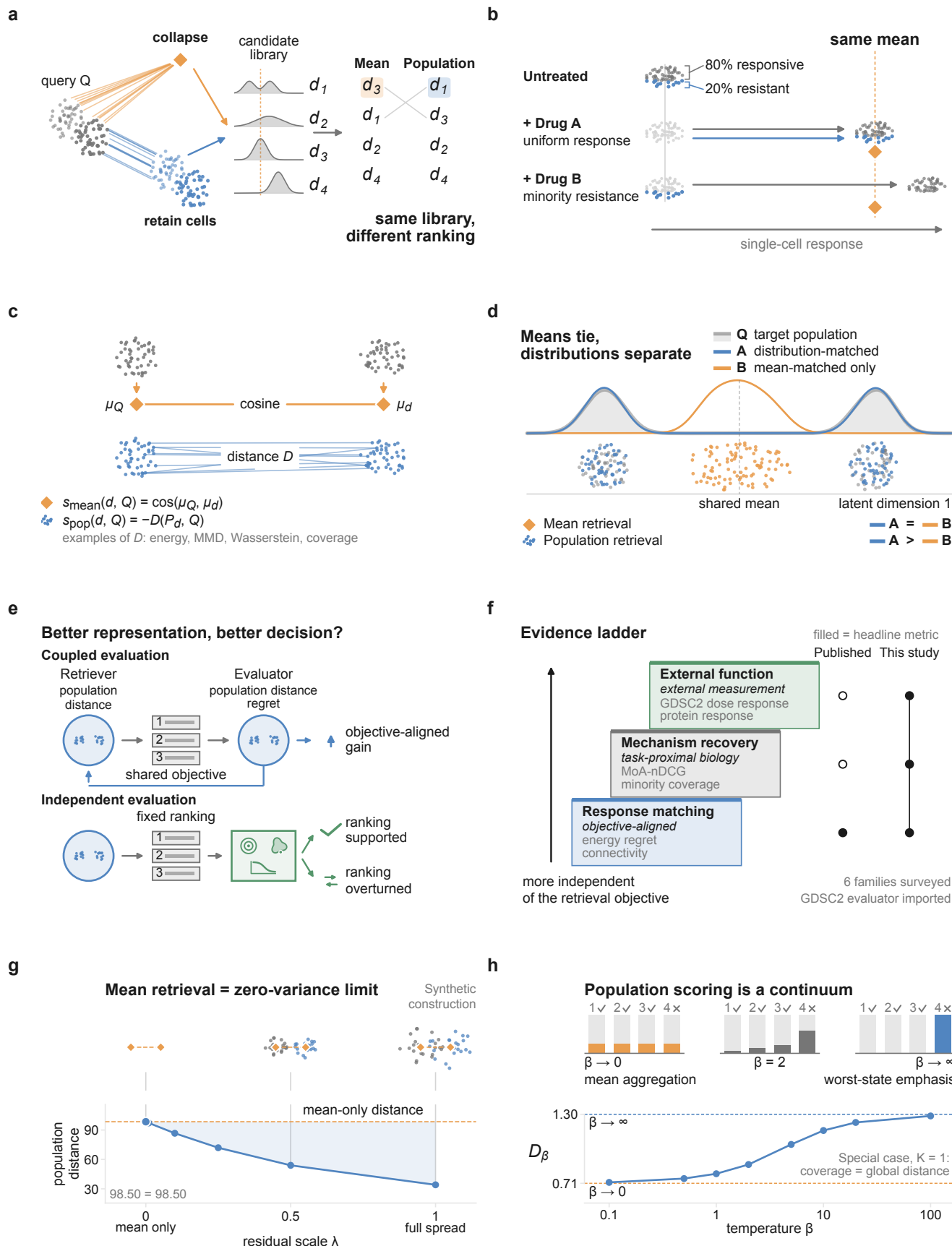


Figure 1: **From perturbation signatures to population-to-population drug retrieval.** Panels **a–f** are schematic and report no measured value; **g** and **h** are seeded synthetic constructions, not measurements on cells. **a**, The task. One query population enters retrieval two ways: collapsed, a fan of leaders carrying every cell into the single mean vector μ_Q (orange diamond), or retained, every cell carried across unchanged. Both routes rank the same library of four candidate populations, drawn as density silhouettes on one shared scale (bimodal, broader, narrow, shifted); the query population is drawn enlarged and is not on that scale. Dashed orange, μ_Q , which is all the collapsed route carries into the comparison. The two ranked stacks are computed from the drawn silhouettes, by distance between means and by 1D Wasserstein distance, and they disagree at rank 1. **b**, Why averaging can fail. The same simulated cells in three lanes, with a faint ghost of the untreated population left at baseline in each treated lane. Under drug A both states travel together and arrive intact; under drug B the responsive majority travels much further and the resistant fifth does not move at all, and stays visible inside the ghost. Both treated population means, orange diamonds, land on one rule: with a fifth of the cells unmoved the responding 80% must overshoot by $1/0.8$, so $0.8 \times 1.25 + 0.2 \times 0 = 1.0$ and the mean cannot separate the two drugs. **c**, What changes mathematically. The same two populations in both rows, cell for cell. The mean route detours below the data, collapsing each population onto a centroid and joining only those two points; the population route joins the cells directly. The two score definitions are set at the foot of the panel behind their route’s own mark, and D is a family of scores rather than one index. **d**, How the ranking changes. Three populations constructed to share one sample mean exactly, with the marginal densities computed from the cells drawn beneath them and one guide through both halves. Candidate A reproduces the target’s bimodal shape, candidate B matches only its mean. A mean score cannot separate them; a population score prefers A. **e**, Why an evaluation can mislead. Coupled, the retriever and the evaluator are the same object drawn the same way, and the path from one to the other closes: the score that produced the ranking is the score that grades it, so the gain it reports is objective-aligned by construction. Independent, the identical fixed ranking is handed to a judge of a different kind, which the retrieval method never saw; the three icons are mechanism of action, target engagement, and GDSC2 dose-response viability. That judge can support the ranking or overturn it. **f**, The evidence ladder. Three levels, drawn as a rising staircase so the ordering survives covering the words, arranged by increasing independence from the retrieval objective and not by strength of claim. Right, how far each body of work reports: filled, that level is its headline metric; open, it is not. Across the six published families surveyed every headline metric is response matching, although four of the six also report a mechanism-level metric secondarily. This study is filled at all three levels, with the external evaluator imported from GDSC2 dose-response data rather than measured here. **g**, Mean retrieval is the zero-variance limit of population retrieval. A response is written $X = \mu + \lambda\epsilon$ with $E[\epsilon] = 0$ and the residual of each of two synthetic populations is scaled by λ ; the three glyphs, hung on the curve by hairlines, are those populations at $\lambda = 0, 0.5$ and 1 . The shaded wash is the gap between the energy distance and the mean-to-mean distance $2\|\mu_P - \mu_Q\|$, which is constant by construction: it opens as spread is restored and closes to nothing at $\lambda = 0$, where the two agree exactly ($98.50 = 98.50$, corner). **h**, Population scoring is itself a continuum. The candidate matches three of the query’s four response states and misses the fourth (cross). Each frame shows the weight the aggregate gives each state, as the filled share of a full track: at $\beta \rightarrow 0$ four equal quarters, at $\beta = 2$ a graded tilt, and at $\beta \rightarrow \infty$ everything on the worst-matched state. The weights are the exact gradient of D_β with respect to the per-state distances, and the middle frame is a measured row rather than an illustration. The curve is D_β between those two analytic endpoints, 0.7125 and 1.300 ; energy distance is exactly the single-partition ($K = 1$) case.

113 Cell populations contain retrievable information beyond mean signatures

114 Having established the connection between mean-signature and population-level retrieval, we next
115 asked whether retaining within-population structure provides additional information for retrieval in
116 practice. We first considered the setting in which candidate responses are directly observed, so that the
117 full cellular populations are available to both retrieval approaches. Across the seven task–setting com-
118 binations analysed here, energy-distance retrieval reached a Hit@1 of 0.837, compared with 0.388 for
119 mean-cosine retrieval and the numerically identical CMap-style cosine baseline^{11,12} (Fig. 2a). The same
120 pattern was observed in PCA space, where population distance reached 0.778 compared with 0.518 for
121 the corresponding mean-based score. These values are unweighted macro-means across task–setting
122 combinations; weighting by the number of queries gave 0.887 for energy and 0.421 for mean cosine
123 (Supplementary Tables 1 and 2).

124 The advantage was substantial on average but not universal. In Frangieh³², the natural biological
125 dataset contributing to this comparison, mean-signature retrieval slightly outperformed energy distance
126 (Hit@1 0.600 versus 0.578; Fig. 2b). Thus, retaining the full response population can reveal information
127 that is lost by averaging, but whether that information improves retrieval depends on the task.

128 We next asked whether this advantage persists when only part of each candidate population is ob-
129 served. In this controlled partial-observation setting, candidates were evaluated using a distribution-
130 sensitive regret criterion that penalizes selecting a response population with poorer latent per-cell out-
131 comes than the best available candidate. Across 765 queries, the worst-case subpopulation-coverage
132 score reduced median regret by +0.118 relative to mean retrieval (mean +0.258), with lower regret on
133 72% of queries (paired Wilcoxon $p = 1.6 \times 10^{-66}$; Fig. 2c). The corresponding median was +0.119 among
134 the 621 queries identified by the pre-specified diagnostic as favourable to population-level retrieval.

135 The signal was not confined to a single distributional score: all five population-level variants showed
136 positive gains under the same response-matching criterion (Fig. 2f). At the same time, simple diagnostics
137 did not reliably identify where those gains would occur. Queries flagged as favourable and unfavourable
138 showed nearly identical median regret reductions (+0.119 and +0.122, respectively; difference -0.003 ,
139 95% bootstrap CI -0.062 to $+0.058$, Mann–Whitney $p = 0.97$; Fig. 2d), and progressively reducing the
140 minority response state from half of the query population to a tenth of it did not produce a consistent
141 monotonic loss of the energy-distance advantage across cell lines (Fig. 2e).

142 These results establish the first part of the problem: single-cell response populations contain retriev-
143 able structure beyond their mean signatures. They do not establish that this additional structure im-
144 proves a biologically meaningful drug ranking. Both source identification and the partial-observation
145 regret criterion reward correspondence between response populations, and therefore remain closely
146 aligned with the information used by population-level retrieval. We next asked whether the same ad-
147 vantage survives when the rankings are judged by biological and functional criteria that are less directly
148 tied to response matching.

149 Additional retrieval information does not necessarily improve biological ranking

150 The preceding analyses show that population-level retrieval can exploit response structure that is lost on
151 averaging. We next asked whether this additional information improves the same candidate rankings
152 when success is defined by biological criteria rather than by response matching itself. To isolate the effect

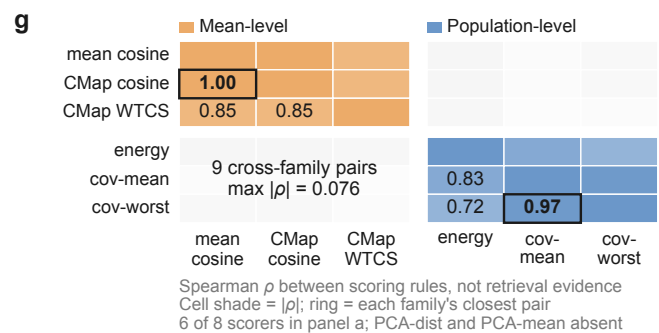
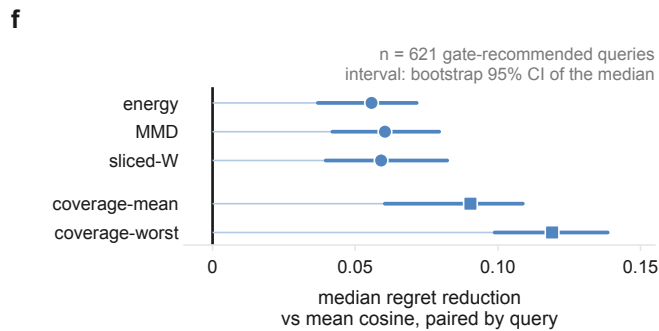
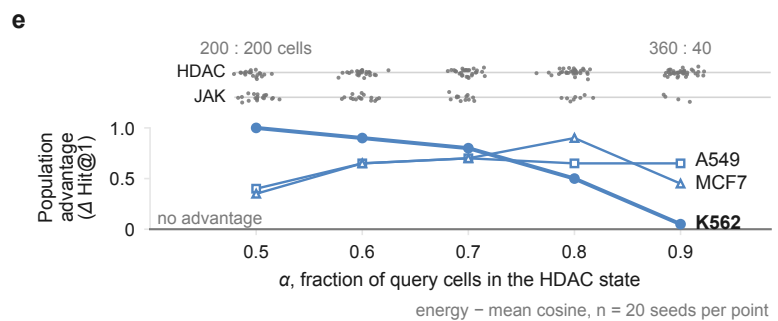
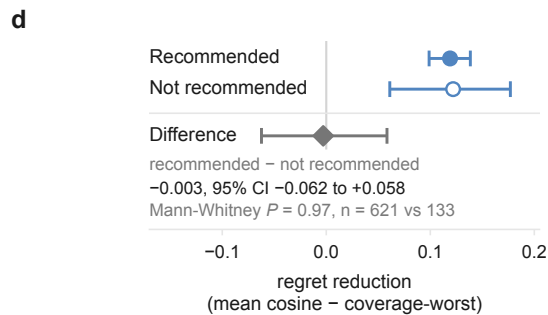
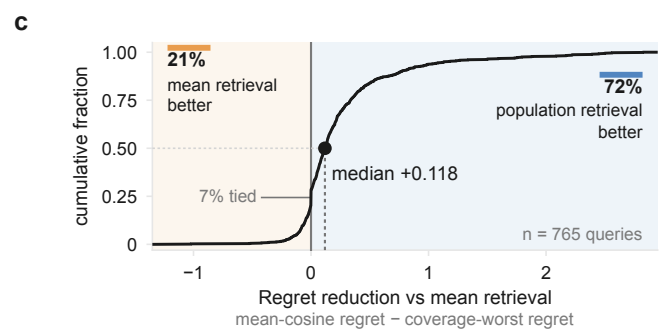
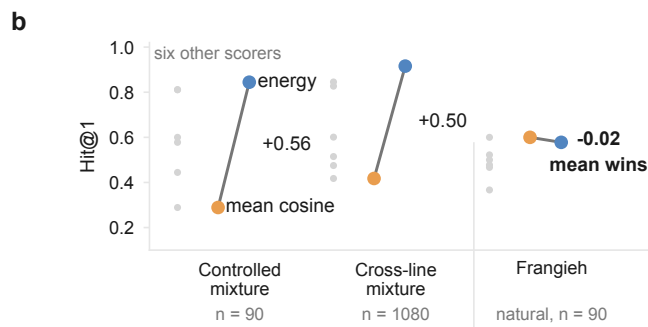
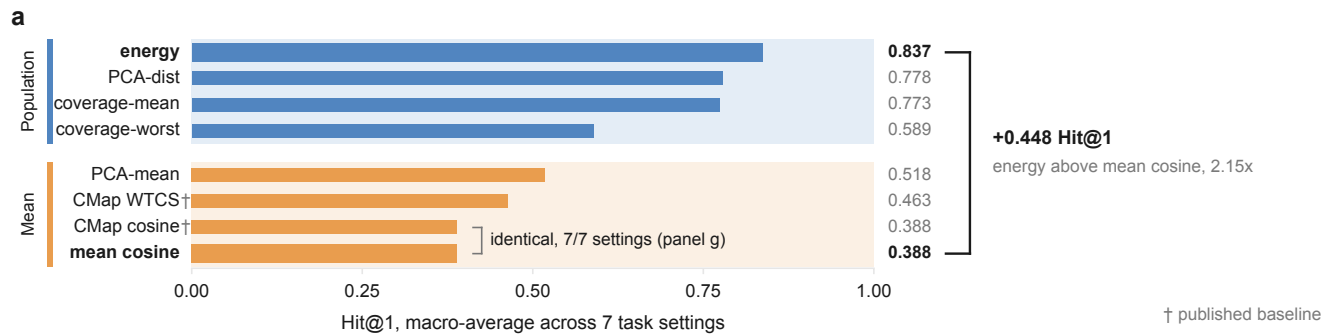


Figure 2: Cell populations carry retrievable information beyond mean signatures. Every criterion in this figure rewards correspondence between response populations and therefore shares the information that population-level retrieval uses; evaluation under criteria that do not is Fig. 3. **a**, Hit@1 for eight retrieval scores, as unweighted macro-means over seven task-setting combinations (controlled $\times$ three cell lines, cross-line $\times$ three line pairs, Frangieh). Scores are grouped by the representation they compare: population-level scores compare retained single-cell populations, mean-level scores compare collapsed per-perturbation signatures. Grouping costs the ordering nothing, because the weakest population-level score (coverage-worst, 0.589) is above the strongest mean-level one (PCA-mean, 0.518). Energy distance reaches 0.837 against 0.388 for mean cosine, a difference of +0.448. Daggers mark published baselines; the implemented CMap-style cosine is numerically identical to mean cosine (both 0.388492; **g**). **b**, Per-task Hit@1 for energy distance and mean cosine, with the six remaining scores as context. The advantage is +0.56 on the controlled mixture and +0.50 on the cross-line mixture, and reverses to -0.02 on Frangieh, the only natural biological dataset entering this comparison, where mean/CMap cosine (0.600) is the best of all eight scores. $n = 90, 1,080$ and 90 queries. **c**, Empirical cumulative distribution of the per-query regret reduction of the worst-case subpopulation-coverage score relative to mean retrieval, over all 765 partial-observation queries. Median +0.118, mean +0.258, the mean pulled up by a right tail reaching +2.82; 72% of queries improve, 21% worsen and 7% are unchanged (paired Wilcoxon $p = 1.6 \times 10^{-66}$). **d**, The same queries split by the pre-specified diagnostic. Median regret reduction is +0.119 among the 621 recommended queries and +0.122 among the 133 non-recommended ones, a difference of -0.003 (95% bootstrap CI -0.062 to $+0.058$; Mann-Whitney $p = 0.97$), so the diagnostic does not enrich for the gain. Points, medians; bars, 95% bootstrap confidence interval; shading, interquartile range. Queries within a cell line share a candidate library, which makes this test anticonservative and therefore strengthens rather than weakens a null result. A further 11 queries flagged mean-sufficient (median exactly 0.000) are not compared; pooling them into the non-recommended group gives the same verdict (difference +0.005, Mann-Whitney $p = 0.81$, interval covering zero). **e**, Population advantage (energy-distance Hit@1 minus mean-cosine Hit@1) against the mixing fraction α of the controlled benchmark, whose 400-cell query is drawn with a fraction α from one mechanism-of-action response state and $1 - \alpha$ from the other, so the minority state falls from 200 cells to 40. The two states are held orthogonal throughout: α changes their proportions, not their similarity. In K562 the advantage falls monotonically from +1.00 to +0.05, whereas in A549 and MCF7 it does not fall (+0.40 to +0.65 and +0.35 to +0.45, neither monotonic). Top, the query composition at each α . $n = 20$ seeds per point, so a single Hit@1 is resolved only to 0.05 and small differences between successive points should not be read. **f**, Median regret reduction for five population-level scores on the 621 diagnostic-positive queries, ordered by construction rather than by value: three global distances between populations, then two subpopulation-coverage variants. All five 95% bootstrap intervals exclude zero. The coverage pair has the larger point estimates, but its intervals overlap those of the global-distance group, so the two sub-families are not separated. **g**, Spearman correlation among six retrieval scores over the 54,180 query-candidate scores generated by 1,260 queries, ordered into the two representation families. Cell shade, $|\rho|$; values printed in the lower triangle; rings, the closest pair within each family. Mean cosine and the implemented CMap-style cosine agree at $\rho = 1.000$ to ten decimal places, which is what makes them one baseline in **a** rather than two; within the population family, coverage-mean and coverage-worst reach 0.97. All nine cross-family correlations satisfy $|\rho| \leq 0.076$. Two of the eight scores in **a**, PCA-dist and PCA-mean, are absent from this export and are not shown. These correlations describe similarity between scoring rules and are not independent statistical evidence for retrieval performance.

153 of the evaluator, we held the query, candidate library, retrieval scorer and resulting ranking fixed and
154 changed only the criterion used to judge that ranking (Fig. 3).

155 We first used mechanism-of-action recovery as a task-proximal biological criterion. Among the
156 600 partial-observation queries for which both response-matching and mechanism annotations were
157 available, the worst-case subpopulation-coverage scorer showed a median improvement of +0.129 over
158 mean retrieval under the response-matching criterion. When the identical rankings were evaluated by
159 mechanism-of-action recovery, however, the median difference was 0.000 and the mean was -0.037
160 (Fig. 3a,b). The change was not driven by a small number of extreme queries: mechanism recovery
161 decreased for 41.3% of queries, increased for 34.2%, and was unchanged for 24.5% (paired Wilcoxon
162 $p = 2.4 \times 10^{-4}$). The same qualitative pattern was observed across all five population-level scoring vari-
163 ants. Thus, the large gain in response matching does not translate directly into improved recovery of
164 biologically related drugs.

165 A second task-proximal criterion asked whether population-level retrieval better preserved minority-
166 state responses. Across 239 real-data retrieval tasks, the mean improvement over mean retrieval was
167 only +0.0018, and in 102 tasks the two approaches selected the same candidate and therefore produced
168 exactly the same score (Fig. 3c,g). The small positive difference was concentrated in the constructed
169 cross-cell-line mixtures (+0.0043), whereas within-line SciPlex3, Frangieh and CD34+ showed gains
170 close to zero. Stratifying queries by their measured response divergence did not reveal a hidden regime
171 of large benefit: the minority-state advantage remained small even in the most divergent queries, and
172 mechanism-of-action recovery showed no significant positive gain in that stratum (Fig. 3c). These results
173 indicate that stronger population differences do not, by themselves, guarantee a more biologically useful
174 ranking (Supplementary Note 2).

175 We next asked whether population-level retrieval carries information that is visible to an external
176 functional readout. For the SciPlex3 drug panel, we constructed a drug-drug functional similarity mea-
177 sure from GDSC2^{30,31} dose-response profiles across 966 cell lines, excluding the three cell lines used
178 to generate the transcriptional queries. After removing the shared cell-line sensitivity effect, energy-
179 distance retrieval reached a Spearman correlation of +0.276 with this external functional similarity, com-
180 pared with +0.083 for the corresponding mean-signature baseline (Fig. 3h). This provides an affirmative
181 result: population-level transcriptional similarity contains functional information that is not captured as
182 strongly by the mean signature (Fig. 3j,k and Supplementary Table 4).

183 Much of that advantage, however, did not require comparing cellular distributions. Ranking candi-
184 dates only by how closely their overall response magnitude matched the query reached +0.232, despite
185 using a single scalar per perturbation. After accounting for this magnitude channel, the remaining as-
186 sociation for energy retrieval was +0.097, positive in all three cell lines but substantially smaller than
187 its unadjusted value (Fig. 3h,i). Energy outperformed the magnitude-only ranking in 62 of 103 queries,
188 but this difference was not significant under the nominal paired comparison ($p = 0.073$), which is itself
189 anticonservative because queries within a cell line share the same candidate pool. Population-level re-
190 trieval therefore carries external functional signal, but much of the signal detected here can be recovered
191 without using cell-to-cell distributional structure.

192 The conclusion is consequently more specific than either a positive or a negative verdict on
193 population-level retrieval. Retaining single-cell populations clearly increases the information available

194 for response matching, and some of that information extends to an external functional readout. But
195 the size—and sometimes even the direction—of the apparent advantage depends on what aspect of
196 the response is used to judge the ranking. This motivates a further question: does using an external
197 readout remove this dependence, or can the statistical form of an external evaluator itself determine
198 which retrieval strategy appears superior?

199 **External evaluation is not automatically neutral**

200 The preceding results show that the apparent value of population-level information changes when a
201 ranking is judged by a criterion different from the one used to construct it. A natural response is therefore
202 to evaluate retrieval against an external measurement that the retrieval algorithm never observes. We
203 asked whether such an external readout is sufficient to provide a neutral judge.

204 Frangieh Perturb-CITE-seq³² provides a direct test because RNA and surface proteins are measured
205 from the same cells, while the retrieval algorithms use only the RNA. We constructed two protein-
206 based evaluators from the same cells and the same protein measurements, differing only in how those
207 measurements were summarized (Fig. 4c). The first compared the mean protein responses of each
208 query–candidate pair. Under this mean-based evaluator, mean-signature retrieval outperformed energy-
209 distance retrieval (+0.242 versus +0.146), consistently across Control, IFN γ and Co-culture conditions.

210 We then evaluated the identical RNA-derived rankings against the full distributions of the same
211 protein responses rather than their means. The ordering reversed. Energy-distance retrieval reached
212 +0.529, compared with +0.334 for mean-signature retrieval, with the same direction in all three immune
213 conditions (Fig. 4c). Neither retrieval method had access to either protein-based evaluator. The reversal
214 therefore does not arise from a method being scored directly by its own training objective. Instead, it
215 shows that even an external and unseen readout can favour a retrieval method whose statistical repre-
216 sentation matches the form in which that readout is evaluated.

217 A response-magnitude control refined this result. A scalar score that compares only the overall mag-
218 nitude of the query and candidate responses rose from +0.125 under the mean-based protein evaluator
219 to +0.352 under the distribution-based evaluator, showing that part of the reversal can occur without
220 comparing cellular distributions at all. Nevertheless, energy distance exceeded this scalar by +0.177
221 under the distribution-based evaluator, compared with only +0.022 under the mean-based evaluator.
222 The protein measurements therefore contain genuine population-level signal, but whether that signal
223 appears advantageous depends strongly on which aspect of the external measurement is preserved by
224 the evaluator.

225 A controlled benchmark illustrates the same dependence under known ground truth (Fig. 4a,b). In
226 HIR-Bench, an analytical boundary separates settings in which subpopulation structure can alter the
227 preferred candidate from those in which the mean is sufficient. However, this synthetic boundary does
228 not transfer reliably to the real datasets analysed here. We therefore use HIR-Bench as a conceptual
229 control for when population structure can matter, rather than as a diagnostic for deciding when it will
230 matter in biological data (Fig. 4g–i and Supplementary Note 1).

231 We next asked whether naturally heterogeneous tissue¹⁷ contains the biological response structure
232 that population-level retrieval is intended to exploit. In glioblastoma samples from ZhaoSims2021, drug
233 responses differed across co-existing cellular compartments: the median cosine between compartment-
234 specific response directions was 0.566 across 17 patient–drug pairs, substantially lower than perfect

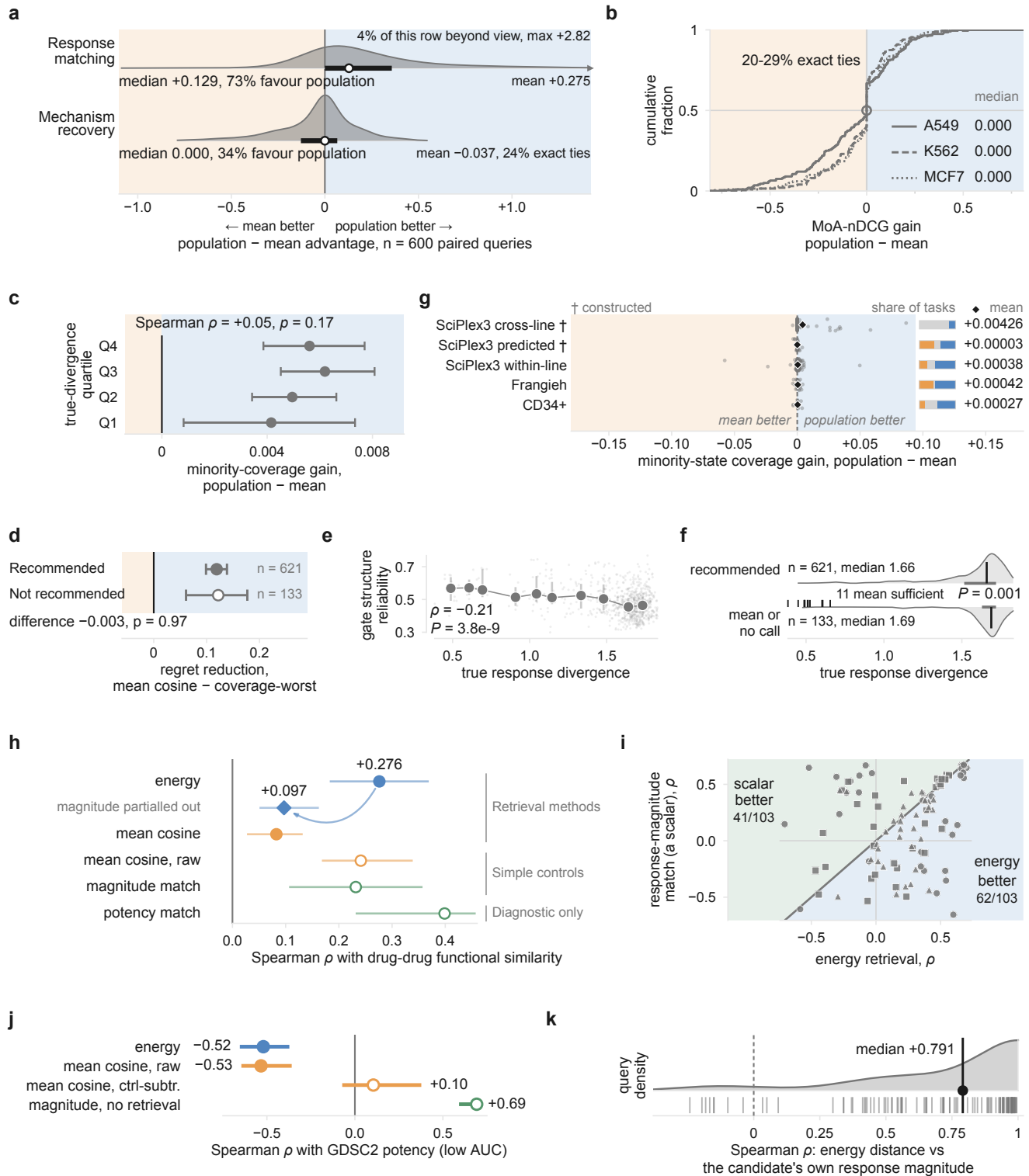


Figure 3: The population advantage weakens, and can change direction, once the judge stops sharing the retrieval objective. Four numbers are the skeleton: +0.129 and 0.000 in **a**, +0.276 and +0.097 in **h**. Points are medians and bars 95% bootstrap intervals unless stated; cell lines are distinguished by marker shape. **a**, All 600 leave-one-drug-out queries and the same rankings, judged twice. Response matching: median +0.129, favouring population retrieval on 73% of queries. Mechanism-of-action recovery: median exactly 0.000, 34% favouring population, 24.5% exact ties (paired Wilcoxon $p = 2.5 \times 10^{-4}$). Means +0.275 and -0.037; 4% of the upper row runs past the view, to +2.82. Nothing was refitted: only the evaluator changed. **b**, The null is distribution-wide, not an average spoiled by outliers. Cumulative distributions per cell line, medians exactly 0.000 in all three ($n = 200$ each); the step at zero is the 20 to 29% of queries selecting the same candidate under both approaches. **c**, Minority-state coverage gain by quartile of measured response divergence ($n = 765$; quartile means). All four are positive and the gain does not grow with divergence (Spearman $\rho = +0.05$, $p = 0.17$; overlapping intervals). The metric’s own median is 0.985 and its interquartile range 0.012; no ratio to a chosen denominator is reported, because that answer ranges from 0.6% to 53% with the denominator. **d**, The pre-specified diagnostic does not concentrate the response-matching gain: +0.119 among 621 recommended queries against +0.122 among 133 non-recommended (difference -0.003, CI -0.062 to +0.058, Mann-Whitney $p = 0.97$). The same measurement as Fig. 2d; **d-f** are three independent ways the diagnostic fails. **e**, Its structure-reliability axis should rise with measured divergence and instead falls (Spearman $\rho = -0.21$, $p = 3.8 \times 10^{-9}$, $n = 765$). Small points, queries; large points, binned medians. **f**, Its thresholded verdict inherits that inversion. The 133 queries declined as “mean or no call” are *more* divergent than the 621 recommended (medians 1.69 against 1.66, Mann-Whitney $p = 0.001$); the 11 declined as “mean sufficient” lie below the fifth percentile of both groups (median 0.51), where the diagnostic is correct. Pooling the two decline verdicts returns $p = 0.09$, which is two opposite effects cancelling rather than an absence (Kolmogorov-Smirnov $D = 0.154$, $p = 0.007$). **g**, Minority-state coverage gain across 239 real-data tasks, one point per task, dataset mean as a diamond, share of tasks above zero at right. The constructed cross-line mixtures have the largest mean (+0.0043) and the *smallest* share above zero (18%, median exactly 0.000); within-line tasks improve on 56% at +0.0004. Overall mean +0.0018, and 102 of the 239 tasks select the same candidate under both. Daggers, constructed data. **h**, External functional evaluation against drug-drug functional similarity from GDSC2 dose-response profiles, the three SciPlex3 cell lines held out ($n = 103$ queries). Filled markers rank candidates; open markers perform no retrieval. Energy retrieval reaches +0.276 against +0.083 for the mean-signature incumbent, but a scalar comparing only response magnitude reaches +0.232, and partialling that channel out leaves energy at +0.097 (arc). Per cell line the scalar-to-energy ratio is 0.95, 1.11 and 0.49, so the pooled comparison is the pre-specified one. **i**, The same comparison per query. Energy exceeds the scalar on 62 of 103 queries; the nominal paired $p = 0.073$ is anticonservative because queries within a cell line share a candidate library. **j**, The same four rankings scored against absolute GDSC2 potency. Positive means the ranking places the most potent candidates first; GDSC2 AUC runs opposite to potency, so the axis is the potency ranking rather than AUC. The two rankings that carry candidate response magnitude change sign (energy -0.52, raw mean cosine -0.53); the control-subtracted mean cosine collapses towards zero (+0.10) without inverting, and a scalar performing no retrieval reaches +0.69. An endpoint diagnostic, not evidence that retrieval fails. **k**, The channel behind **h** and **j**: the per-query Spearman correlation between the query-candidate energy distance and the candidate’s own response magnitude, median +0.791. A left tail reaches below zero, so this describes the typical query and not every one.

alignment and therefore consistent with state-dependent drug responses (Fig. 4e). This heterogeneity was also recoverable in principle. For within-patient drug-response partitions, a supervised probe reached an accuracy of 0.923, whereas the best unsupervised procedure reached 0.777, leaving a median gap of +0.117 (Fig. 4d). In real tissue, the relevant information is therefore present, but current unsupervised recovery does not fully extract it (Supplementary Note 4).

The remaining question is whether this additional structure is large enough to change which drug should be preferred. Here the tumour data provide a constraint rather than a retrieval result. Across 18 within-patient drug pairs, similarity between mean drug-response signatures strongly tracked similarity between malignant-cell responses. Because these quantities partly share cells, we repeated the analysis using disjoint compartments: myeloid-response similarity still ranked malignant-response similarity at Spearman $\rho = +0.835$ (Fig. 4f). Thus, drugs that look similar at a broader population level also tend to look similar within a specific cellular compartment, leaving less room for within-population structure to reorder candidates than the existence of heterogeneity alone would suggest. No retrieval was performed on this patient cohort, so these analyses do not demonstrate the presence or absence of a population-level retrieval advantage. Instead, they identify the missing requirement that the next section addresses: additional cellular structure must not only exist and be recoverable, but must also be relevant enough to change candidate preference.

When can population-level information improve drug retrieval?

The preceding results suggest that retaining single-cell populations is useful only when the information they preserve can propagate all the way to candidate ranking. We therefore separated this problem into two measurable information requirements and a third, downstream requirement. First, different cellular states must respond differently to the same perturbation. Second, those differences must be recoverable from the observed response population. Even when both conditions hold, however, the additional information can improve retrieval only if it is large enough to change which candidate is preferred (Fig. 5a). We refer to these requirements as differential response, recoverability and decision relevance.

We first examined differential response in the candidate populations supplied to the retriever. When candidate responses were generated by perturbation predictors rather than directly observed, the advantage of population-level retrieval became small and inconsistent: the change in nDCG@10 relative to mean retrieval was -0.016 for an average-effect predictor, -0.071 for nearest-neighbour transfer and $+0.027$ for a linear-latent predictor (Fig. 5b). The limiting case is simple. Any predictor that adds the same perturbation effect to every cell in a context gives every cellular state the same response vector and therefore cannot generate state-specific differential response. This is an algebraic property of the model class rather than a consequence of insufficient training (Fig. 5h-j and Supplementary Note 3).

A within-context analysis confirmed that this distinction is biologically relevant. Real treated cells, partitioned by their pre-perturbation state, produced markedly different response directions, with a mean induced-response cosine of 0.205 across 30 cell-line and drug combinations. By contrast, the additive average-effect and linear-latent predictors returned a cosine of 1.000 by construction, while a deployed scGen model⁶ remained close to this additive limit at 0.972. A non-parametric optimal-transport map^{22,33} moved further away from the additive limit but remained substantially more aligned than the observed responses (0.900, s.d. 0.107; Supplementary Note 3). Thus, preserving the heterogeneity of the

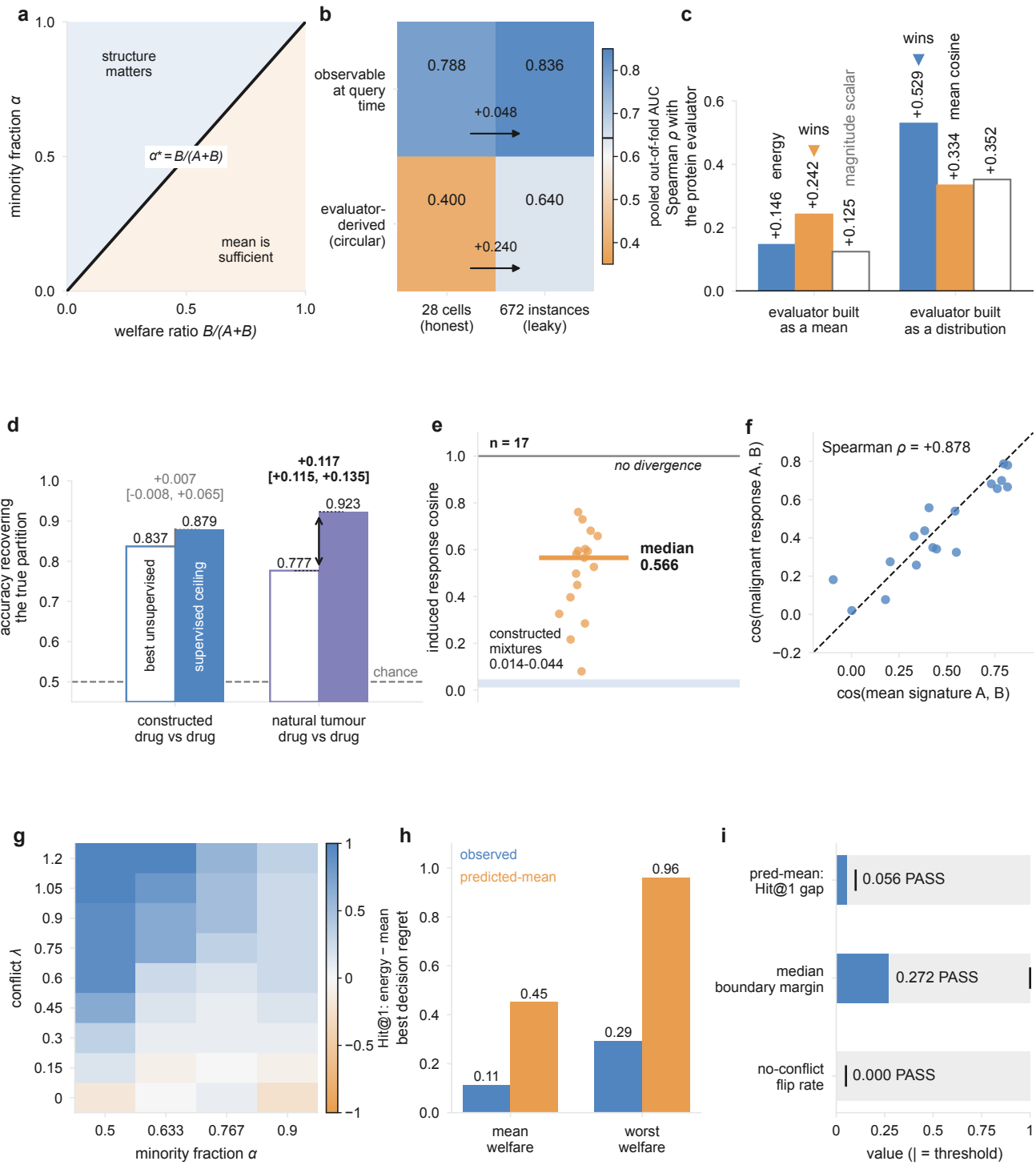


Figure 4: External evaluation remains sensitive to statistical form, while natural tissue constrains the value of population structure. Filled marks are scores that compare a response representation; open marks are controls that perform no retrieval. **a–c**, Analytical boundary, synthetic benchmark and two real evaluators. **a**, In HIR-Bench the analytical boundary $\alpha^* = B/(A + B)$ separates regimes where the mean is sufficient from those where subpopulation structure can change the preferred candidate. **b**, Circularity measured rather than argued: pooled out-of-fold AUC for predicting retrieval failure, over two feature sets by two cross-validation units. Features observable at query time reach 0.788 on the honest 28-cell unit and 0.836 on the leaky 672-instance unit; features derived from the benchmark’s latent utility construction reach 0.400 and 0.640 across the same two units, so the leakage alone is worth +0.048 and +0.240 (arrows). The rule on the colour bar marks the 0.643 majority-class rate. The 0.400 is not itself interpretable as below-chance skill: with 28 label-determining parameter cells every held-out fold is single-class, so no per-fold distribution exists. The synthetic boundary is therefore a conceptual control, not a diagnostic for real data. **c**, Two external protein-based evaluators built from the same cells and the same protein measurements ($n = 659$ queries), differing only in how those measurements are summarized. Energy distance compares protein-response cell clouds; mean cosine is the pre-specified incumbent; the response-magnitude scalar is a control that compares no distributions. Bars appear in that order in both columns. Mean-based evaluator, mean-signature retrieval outperforms energy distance (+0.242 versus +0.146); distribution-based evaluator, the ordering reverses (+0.334 versus +0.529), in all three immune conditions. The scalar rises from +0.125 to +0.352 between the two evaluators, yet energy still exceeds it by +0.177 under the distributional one. **d–f**, The same question in naturally heterogeneous tumour tissue. **d**, Recoverability of within-patient drug-response structure in glioblastoma; the natural comparison is within one compartment of one patient. Open bars, best unsupervised method; filled bars, supervised probe (0.923 versus 0.777; median gap +0.117, 95% CI +0.115 to +0.135, patient-level cluster bootstrap, $n = 36$ splits against 24 in the constructed arm), so the information is present but not fully recovered without supervision. The gap is a median of paired per-split differences, not the difference of the plotted medians, and survives two checks: fixing every probe–clusterer pair gives +0.122 in tissue against –0.002 in the constructed arm, and dropping the patient that contributes 30 of 36 splits gives +0.123 (+0.098 to +0.138). **e**, Cellular compartments in the same tumour respond differently to the same drug: median cosine between compartment-specific response directions 0.566 across 17 patient–drug pairs from ten donors. The rule at the top marks cosine = 1, complete alignment; the shaded band at the foot is the range spanned by our own constructed mixtures, which are near-orthogonal by construction. **f**, Mean-level drug similarity nevertheless tracks compartment-specific response similarity: across disjoint compartments, myeloid-response similarity ranks malignant-response similarity at Spearman $\rho = +0.835$ ($n = 18$ drug pairs, 15 from one patient; dashed line, equal similarity). No retrieval was performed on this cohort; the analysis tests a mechanism rather than measuring a retrieval advantage. **g–i**, HIR-Bench’s own verification, under the boundary in **a** and the leakage test in **b**. **g**, Difference in Hit@1 between population-level and mean retrieval across the subpopulation-balance and preference-conflict grid, $n = 12$ seeds per cell. The distributional advantage is regime-dependent: it runs from –0.25 at zero conflict and $\alpha = 0.9$ to +1.00 in three high-conflict cells. The colour scale is fixed at ± 1 because the quantity is a difference of two Hit@1 rates and is bounded there by construction, so the pale field is the statistic’s whole possible range rather than a stretched one. **h**, Decision regret under fully observed populations and under predicted populations retaining only mean-level response information. Each bar is the best value over nine retrieval methods, each averaged over 72 runs, so the four bars compare best-of envelopes rather than one method across conditions. Collapsing candidates to their means increases regret most strongly when the criterion depends on the poorer-served subpopulation. **i**, Internal benchmark verification against three thresholds (marks): the no-conflict flip rate is zero against 0.05, as required by the generator; the median boundary margin is 0.272 against 1.0; and under the predicted-mean information condition population-level and mean-signature retrieval agree, with a Hit@1 gap of 0.056 against 0.10. HIR-Bench is a constructive demonstration that within-population structure can change candidate preference, not a discovered biological law.

starting population is not sufficient: a predictor must also reproduce how the perturbation acts differently across cellular states. We did not identify a predictor in these experiments that both transferred accurately across contexts and reproduced this differential response, so whether opening this information channel is sufficient to improve cross-context retrieval remains unresolved.

Differential response is also not sufficient if the relevant structure cannot be recovered from the observed population. We tested recoverability in a controlled two-state mixture with known labels. The best unsupervised method reached an accuracy of 0.674, but the supervised ceiling was only 0.692, indicating that little additional recoverable information remained in this constructed representation (Fig. 5e). This apparent ceiling was partly a consequence of pooling multiple drugs into broad classes: posing the same problem at the level of individual drug pairs raised the supervised ceiling to 0.879. Natural tissue showed a different limitation. As shown in Fig. 4d, glioblastoma responses were highly decodable under supervision (0.923) but less well recovered by unsupervised methods (0.777), indicating that recoverability can become an algorithmic rather than an informational bottleneck.

The final requirement is that the recovered information must alter the candidate ordering itself. This condition is not directly measured as the first two are, but several analyses constrain it. Stratifying the retrieval queries by their measured response divergence did not reveal a regime in which stronger state-specific responses produced a clear mechanism-of-action advantage: even the most divergent quartile showed no significant positive gain (mean $+0.003$, $q = 0.57$), whereas the lowest-divergence quartile was significantly negative (Fig. 5g). Similarly, the tumour analysis showed that compartment-specific drug responses can differ while preserving much of the ordering already present at the broader population level (Fig. 4f). Thus, heterogeneity can be biologically real and recoverable without being large enough to reorder candidate drugs.

The same pattern was reproduced at larger scale in Tahoe-100M¹⁶. Across 3,630 drug-cell-line conditions, the median cosine between state-specific response directions was 0.739, with fewer than 0.11% of conditions as divergent as the constructed mixtures used earlier. The recoverability gap nevertheless remained measurable across 960 drug pairs and 48 cell lines (median gap, 0.157), while a disjoint comparison of state-specific response ordering remained high ($\rho = 0.841$ across 44 evaluable contexts). These measurements reinforce the distinction between the three requirements: differential responses can exist, and can be recoverable, while still leaving limited room for population structure to change candidate preference (Fig. 5k–n and Supplementary Note 4).

Taken together, these results define when population-to-population retrieval can plausibly add value beyond a mean signature. The candidate responses must contain state-specific perturbation effects; those effects must be recoverable from the measured population; and, ultimately, they must carry information that changes the ranking of candidate drugs. The first two requirements determine whether useful population-level information is available. The third determines whether that information matters for the retrieval decision.

Discussion

Single-cell perturbation atlases^{14–16} change the representation available for drug retrieval: a perturbation can now be described by a population of cellular responses rather than by a single averaged signature. Our results show that this added resolution matters, but not simply because preserving more cellular

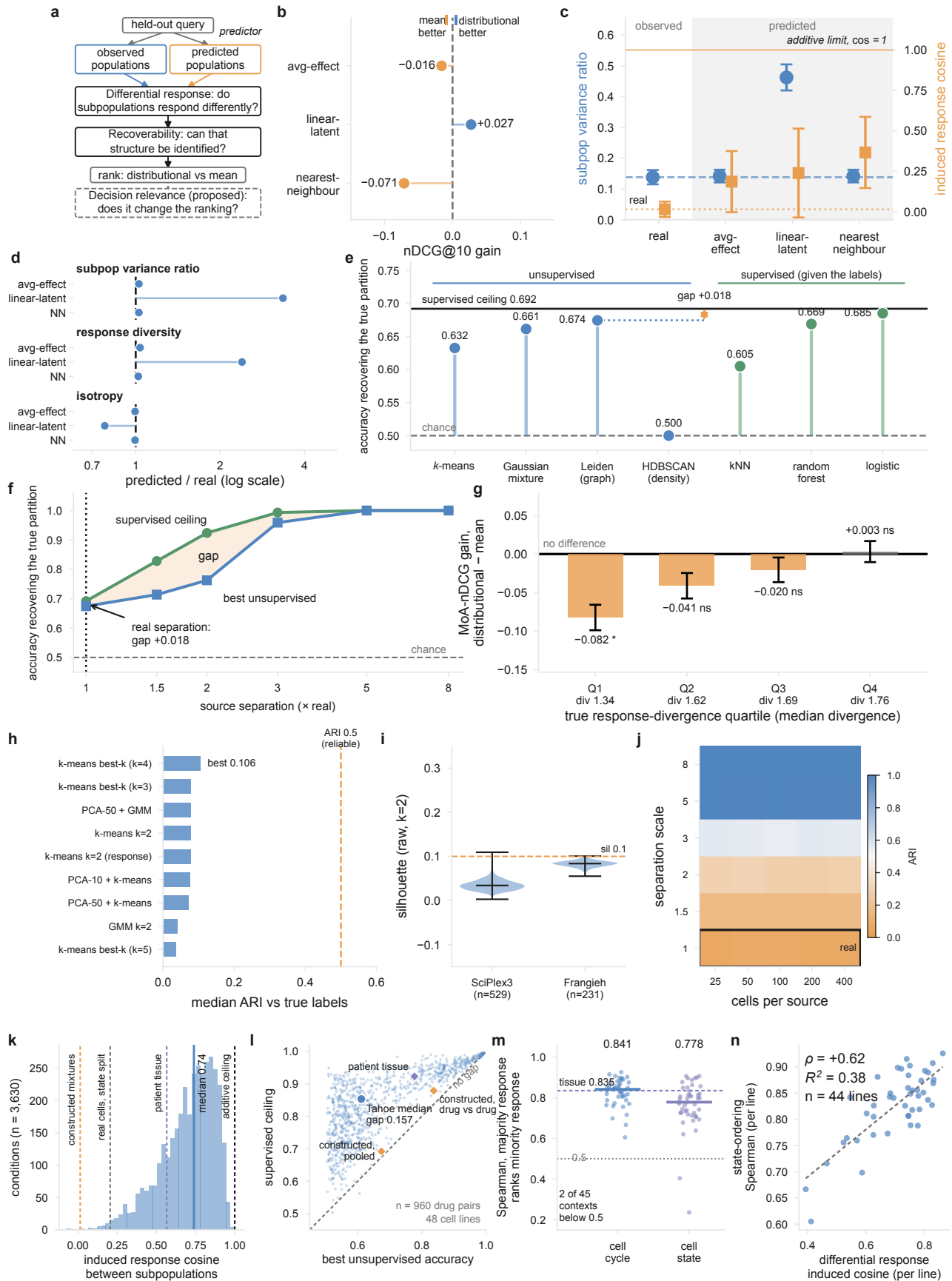


Figure 5: When can population-level information improve drug retrieval? **a**, Three requirements connect cellular heterogeneity to a change in candidate ranking. Differential response and recoverability are measured directly in this study; decision relevance is a downstream requirement proposed from the combined results and is shown separately. **b**, When candidate populations are generated by perturbation predictors rather than directly observed, the population-level advantage is small and inconsistent across the three predictor families tested (distributional minus mean nDCG@10: -0.071 , -0.016 and $+0.027$). **c**, Differential response. Predicted populations can preserve heterogeneity in the starting cell population while failing to reproduce state-specific perturbation effects; additive predictors necessarily assign the same perturbation response to every cellular state. **d**, Structure diagnostics of predicted candidate populations relative to the real population they replace, on a ratio axis with the reference at 1. The additive average-effect and nearest-neighbour predictors reproduce subpopulation variance, response diversity and isotropy almost exactly (ratios 0.99 – 1.04); the linear-latent model overshoots the first two (3.35 and 2.39) and reduces isotropy (0.78). **e**, Recoverability in a controlled two-state mixture. The best unsupervised method reaches 0.674 against a supervised ceiling of 0.692 ; the low ceiling is partly explained by pooling multiple drugs into broad classes, whereas individual drug pairs reach a ceiling of 0.879 . Natural tumour tissue shows a larger supervised–unsupervised gap (Fig. 4d). **f**, Increasing the separation of the constructed states pulls the supervised ceiling away from unsupervised recovery, confirming that the probe can detect additional recoverable information when it is present. Shading marks the region unsupervised clustering cannot reach. **g**, Decision relevance is not implied by differential response alone. Mechanism-of-action gain does not become a significant positive advantage in the highest response-divergence quartile (mean $+0.003$, $q = 0.57$), whereas the lowest-divergence quartile is significantly negative. Asterisk, $q < 0.05$ after Benjamini–Hochberg correction; ns, not significant. **h–j**, Recoverability measured three further ways on the constructed mixture of **e**. **h**, Unsupervised recovery of the known two-source structure across clustering methods and random seeds; no configuration reaches the 0.5 reliability line. The three best- k rows are one method, k -means with k chosen from 3 , 4 and 5 by silhouette, split by which k won. **i**, Silhouette scores computed independently for perturbation-response populations in SciPlex3 ($n = 529$) and Frangieh ($n = 231$). Most populations separate weakly under a two-cluster partition: visible multimodality is not a prerequisite for, and should not be equated with, recoverable state-specific perturbation information. **j**, Seed-averaged recovery across source separation and cell number. Increasing separation improves recovery, whereas increasing the cell budget alone has little effect over the range examined. The separation transform was defined from the full source populations before subsampling. **k–n**, All three requirements measured on Tahoe-100M plate 3: 50 cancer cell lines, 93 compounds at one concentration and a matched vehicle, giving unconstructed within-cell-line heterogeneity at far larger scale than the other datasets here. No retrieval was run on Tahoe-100M. **k**, Differential response between two cellular states. Across 3,630 evaluable drug–cell-line conditions the median cosine between state-specific response directions is 0.739 ; fewer than 0.11% of conditions are as divergent as the constructed mixtures used in the controlled analyses (dashed anchors). **l**, Recoverability of drug-response structure. Across 960 drug pairs and 48 cell lines the supervised ceiling is 0.853 and the best unsupervised method reaches 0.611 , a median paired gap of 0.157 that is positive in all 48 evaluable cell lines. **m**, Disjoint state-ordering agreement, used as a proxy for the headroom available for state-specific information to alter candidate ordering. Median Spearman correlation between drug–drug similarity matrices computed from non-overlapping cellular states is 0.841 (cell-cycle partition) and 0.778 (independently control-derived partition). Lower values indicate greater scope for reordering; this is not a direct retrieval-performance measure. **n**, Differential-response alignment against disjoint state-ordering agreement across contexts ($\rho = +0.62$, $R^2 = 0.38$, $n = 44$). The two measurements are related but not equivalent, distinguishing the existence of state-specific responses from whether they are large enough to alter candidate ordering. Both partitions were specified independently of downstream performance.

316 detail produces a better ranking. Population-level retrieval recovers information that is lost by averaging,
317 ing, yet whether that information improves a ranking depends on how success is defined. Large gains
318 under response matching can disappear under mechanism-of-action evaluation, and even external read-
319 outs can favour opposite retrieval strategies depending on whether they are summarized as means or
320 retained as distributions. The central distinction is therefore not mean versus distributional retrieval,
321 but information that is available versus information that changes the decision. Classical perturbation
322 signatures and single-cell populations are thus better viewed as points on a continuum of retained re-
323 sponse structure; the relevant question is how much of that structure must be preserved for the decision
324 at hand.

325 This distinction has direct implications for artificial intelligence virtual cells^{1,2}. A virtual cell that
326 accurately predicts an average perturbation response may still fail to provide the information needed
327 for population-level intervention selection. In particular, a model that applies the same perturbation
328 effect^{6,7} to every cell can preserve heterogeneity in the starting population yet, by construction, cannot
329 generate state-specific differential responses. A downstream population-aware retriever cannot recover
330 variation that the predictor never generated. Models intended to support intervention selection must
331 therefore do more than predict mean responses: they must reproduce how perturbation effects vary
332 across cellular states, and those differences must remain recoverable in the predicted population. We
333 could not test whether satisfying this requirement is sufficient to improve cross-context retrieval because
334 no predictor available here both transferred accurately across contexts and reproduced the observed
335 degree of differential response. This gap defines a concrete target for virtual-cell modelling rather than
336 a general verdict on existing model families.

337 Taken together, the results suggest three conditions under which population-level retrieval can add
338 value beyond a mean signature. Cellular states must respond differently to a perturbation, those differ-
339 ences must be recoverable from the measured or predicted population, and the additional information
340 must be consequential enough to change which candidate is preferred. The first two determine whether
341 useful population-level information is available; the third determines whether it matters for the retrieval
342 decision. This final requirement also clarifies what constitutes convincing validation. An external mea-
343 surement is not automatically a neutral judge if its statistical form preferentially preserves the same
344 structure as the retrieval method^{21,29}, so validation should ask whether the added cellular information
345 changes an outcome relevant to the intended use and should compare against simple alternatives such
346 as response magnitude²⁷ before attributing a gain to population structure itself. The functional readouts
347 available here remain indirect and do not establish therapeutic utility. More broadly, richer cellular mea-
348 surements justify richer retrieval methods only when the information they add is retained, recoverable
349 and consequential for intervention choice.

350 **Methods**

351 **Study design and data resources**

352 This study asks when retaining the full population of single-cell perturbation responses improves re-
353 trieval beyond a conventional mean expression signature. The analyses were therefore organized around
354 four distinct roles: observed-population retrieval, evaluation against biological and functional criteria,
355 measurement of naturally occurring response heterogeneity, and large-scale assessment of the condi-

tions under which population-level information can be useful. These roles were analysed separately rather than pooled into a single benchmark.

The primary observed-population retrieval analyses used SciPlex3¹⁵, comprising 276,325 cells across three cancer cell lines (A549, K562 and MCF7), 188 compounds shared across the three contexts and a 2,000-gene expression space. SciPlex3 supports both within-context retrieval and controlled cross-context constructions because the same compound library is measured across all three cell lines. Frangieh melanoma Perturb-CITE-seq³² contains 218,023 RNA-profiled cells across Control, IFN γ and Co-culture conditions with 239 CRISPR knockouts and 2,000 highly variable genes. We use this dataset as an independent perturbation-retrieval setting and, because matched surface-protein measurements are available, for the external evaluator-shape experiment described below. It is a genetic-perturbation test of the same retrieval formulation rather than a drug-retrieval benchmark. CD34+ haematopoietic progenitor data (GSE306429)³⁴ contain 33,984 cells and 36 compounds and are used as a negative-control setting for population-level retrieval.

Three additional resources serve roles outside the primary retrieval benchmark. ZhaoSims2021¹⁷ contains drug-treated patient-derived glioblastoma tissue and is used to test whether state-dependent responses and their recoverability are present in naturally heterogeneous tissue; no retrieval performance is reported from this cohort. Tahoe-100M¹⁶ is used as a larger-scale measurement of the same information requirements under intrinsic within-cell-line heterogeneity. We analyse plate 3 only, which comprises 4,158,278 cells across 50 cell lines and 93 compounds, using the released 2,304-gene log1p-transformed representation without refitting the preprocessing pipeline. Finally, GDSC2 dose-response profiles provide an external functional readout for drug-drug similarity and are joined to the SciPlex3 compound panel independently of the transcriptomic retrieval scores.

Observed responses, predicted responses, external functional measurements and natural-tissue analyses therefore answer different questions in this study. Results are not combined across these resources as though they represented independent replicates of one benchmark. Dataset scale, inclusion criteria, experimental role and query construction are summarized in Supplementary Table 1.

Preprocessing and construction of perturbation-response populations

Single-cell expression analyses were performed in log-transformed expression space using the processed gene sets defined for each dataset. SciPlex3, Frangieh and CD34+ were analysed in 2,000-gene representations. For SciPlex3 and Frangieh these genes were selected in this study rather than inherited from the released objects: counts were normalized to a total of 10^4 per cell and log1p-transformed, 2,000 highly variable genes were chosen by the Seurat criterion³⁵ on a subsample of cells, and the final matrices were then rebuilt from raw counts on those genes using all-gene size factors so that normalization is applied once. SciPlex3 was additionally downsampled to at most 120 cells per compound, cell line and dose, and 3,000 control cells per cell line. The CD34+ tensor was received in processed form and its feature-selection command is not recorded in this repository. For ZhaoSims2021, counts were normalized to a total of 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes. Tahoe-100M was used in the released preprocessed form (2,304 highly variable genes, log1p transformed), with no additional feature selection or representation fitting. Dataset-specific filtering and minimum-cell requirements for individual analyses are given in the corresponding Methods sections below.

For a perturbation d measured in cellular context c , let

$$X_{d,c} = \{x_1^{(d,c)}, \dots, x_{n_{d,c}}^{(d,c)}\}$$

denote its treated single-cell expression profiles and let

$$\mu_{0,c} = \frac{1}{n_{0,c}} \sum_{j=1}^{n_{0,c}} x_j^{(0,c)}$$

denote the mean of the matched control population. The conventional perturbation signature is the control-referenced mean response

$$\bar{d}_{d,c} = \frac{1}{n_{d,c}} \sum_{i=1}^{n_{d,c}} x_i^{(d,c)} - \mu_{0,c}.$$

396 Mean-signature retrieval operates only on $\bar{d}_{d,c}$.

Population-level retrieval retains the corresponding control-referenced cellular responses,

$$P_{d,c} = \left\{ x_i^{(d,c)} - \mu_{0,c} \right\}_{i=1}^{n_{d,c}},$$

397 so that the mean of $P_{d,c}$ is exactly the conventional mean perturbation signature while variation among
398 its cells remains available to the retrieval score. This construction makes the comparison between mean-
399 signature and population-level retrieval differ in the response structure retained rather than in the un-
400 derlying perturbation or control definition.

401 Controls were matched to the biological context in which the perturbation was measured. For anal-
402 yses that explicitly compare responses between cellular states or compartments, each state is instead
403 referenced to its own matched control population so that pre-existing differences between states can-
404 not be mistaken for differential perturbation responses. This state-specific construction is used for the
405 differential-response analyses in predicted populations, patient-derived glioblastoma and Tahoe-100M
406 and is defined in the corresponding sections below.

407 Cell sampling was performed only after the response populations had been defined. Minimum cell
408 numbers, per-condition caps and random seeds differ by experiment and are reported with the corre-
409 sponding analysis rather than imposed globally. Protein measurements used for the evaluator-shape
410 experiment were processed separately, including barcode-based RNA-protein matching and protein-
411 specific normalization, as described below.

412 Population-to-population retrieval and signature baselines

413 For a query response population Q and a library of candidate response populations $\{P_d\}_{d=1}^D$, retrieval
414 assigns each candidate a similarity or distance to the query and orders the candidate library accord-
415 ingly. Population-level methods operate directly on the empirical response populations defined above,
416 whereas signature-based methods first reduce each population to its control-referenced mean response.
417 Query construction and candidate libraries are kept identical when comparing the two representations.

418 **Population-level retrieval.** Five population-level variants were evaluated: energy distance, MMD,
419 sliced Wasserstein, coverage-mean and coverage-worst. Energy distance³⁶ was the primary score. The

420 RBF-kernel maximum mean discrepancy (MMD)³⁷ uses a median-heuristic bandwidth evaluated at
 421 three scales (0.25, 1 and 4 times the median squared pairwise distance of the pooled sample), because
 422 bandwidths fixed in advance underflow in a 2,000-dimensional gene space. The sliced Wasserstein-1 dis-
 423 tance³⁸ averages quantile-matched one-dimensional projections over 128 random directions, which ac-
 424 commodates unequal sample sizes. The final two variants are the two limits of the single subpopulation-
 425 coverage family defined below. All five rank candidates from smallest to largest distance to the query.
 426 Candidate populations were subsampled to a fixed maximum cell count with a fixed random seed be-
 427 fore any distance was computed (500 cells by default; 300 in the partial-observation experiments); a
 428 population left with fewer than two cells was assigned a fixed large penalty distance rather than silently
 429 dropped.

Subpopulation-coverage score. The query is partitioned into K components by k -means on the query cells ($k = 2$ throughout, three initializations, fixed seed), and each candidate cell is assigned to the component whose query centroid it most resembles in cosine similarity. Writing d_k for the base distance between the query cells and the candidate cells of component k , the coverage distance and its ranking score are

$$D_\beta = \frac{1}{\beta} \left(\log \sum_{k=1}^K e^{\beta d_k} - \log K \right), \quad S_\beta = -D_\beta.$$

430 The base distance d_k is one of energy, MMD or sliced Wasserstein, so coverage is a family built on the
 431 same kernels rather than an independent metric. A candidate that populates no cell of a query compo-
 432 nent receives the penalty distance for that component, which is how the score charges a candidate for
 433 leaving part of the query uncovered.

434 Three properties fix the interpretation of β . As $\beta \rightarrow 0$, $D_\beta = \text{mean}(d) + \frac{\beta}{2} \text{var}(d) + O(\beta^2)$ and the
 435 aggregate becomes the arithmetic mean over components; the implementation evaluates this expansion
 436 directly at small β , because the $(\log \text{sumexp} - \log K) / \beta$ form cancels catastrophically there. As $\beta \rightarrow \infty$,
 437 $D_\beta \rightarrow \max_k d_k$, the worst-matched component. For $K = 1$, $D_\beta = d_1$ for every β , so the coverage family
 438 contains the global distance exactly rather than as an approximation. Coverage-mean and coverage-
 439 worst are these two limits, and are the two variants used in the main analyses.

440 **Mean-signature baselines.** For each response population $P_{d,c}$, the corresponding mean signature is
 441 $\bar{d}_{d,c} = \text{mean}(P_{d,c})$. Mean-signature retrieval ranks candidates using either cosine similarity or Euclidean
 442 distance between these vectors. The implemented CMap-style cosine baseline applies cosine similarity to
 443 the same control-referenced mean differential-expression vectors and is therefore numerically identical
 444 to mean-cosine retrieval. The canonical L1000 weighted connectivity score (WTCS) is included as a
 445 separate signature-based baseline. WTCS uses rank-based enrichment of the mean-collapsed expression
 446 response and is related to, but not algebraically identical to, the cosine implementation used here.

447 To test whether the comparison depended on the original gene space, we also evaluated a PCA rep-
 448 resentation. PCA was fitted separately for each retrieval query, after which the same comparison was
 449 performed either between latent-space means or between the corresponding latent response popula-
 450 tions.

Connection between mean and population retrieval. To formalize the relationship between the two representations, we write a response population as

$$X = \mu + \lambda\epsilon, \quad E[\epsilon] = 0,$$

where μ is the mean response, ϵ represents within-population variation and λ controls its magnitude. As $\lambda \rightarrow 0$, the response population collapses to a point mass at μ , and population comparison therefore loses all information beyond the mean. Mean-signature retrieval is the mean-only endpoint of this variance-sensitivity construction.

This limiting argument does not imply that every population distance becomes algebraically identical to every conventional signature score. The exact identity used in this study is narrower: mean-cosine retrieval and the implemented CMap-style cosine baseline apply the same cosine operation to the same mean differential-expression vectors. WTCS remains a distinct rank-based signature score, and energy, MMD and sliced Wasserstein retain their own distance definitions. The continuum therefore refers to the amount of within-population response structure available to the scorer, not to an equivalence between otherwise different scoring functions.

Response-matching and partial-observation evaluation

Observed-population retrieval. When complete candidate response populations were available, retrieval was evaluated as source identification. For each query, the candidate library contained the perturbation that generated the query response together with competing perturbations defined by the corresponding task. Hit@1 is the fraction of queries for which the true source perturbation is ranked first. Comparisons between population-level and mean-signature retrieval use the same queries and candidate libraries. Aggregate Hit@1 values are reported as unweighted macro-means over the seven task-setting combinations used in the observed-population analysis; query-weighted averages are reported as a sensitivity analysis so that the largest task does not determine the primary summary.

Partial-observation protocol. We additionally evaluated retrieval when only part of each candidate response population was available to the retriever. This setting was designed to separate the information used to construct a ranking from the criterion subsequently used to evaluate that ranking. For each query, the population-level and mean-signature methods rank the same candidate library once; the resulting rankings are then retained unchanged while different evaluation criteria are applied.

Three splits construct the partial observation, all on SciPlex3 and all leakage-safe³⁹ in the sense that the hidden material never enters the candidate library. In *leave-drug-out*, one drug is hidden, the query is that drug’s own treated population (up to 200 cells) and the library is every other drug in the context. In *leave-MoA-out*, an entire annotated mechanism class is removed from the library and the query is one drug drawn from the removed class, so recovery requires a same-target surrogate among the remaining drugs. In *partial-library*, a fraction of the drugs is treated as measured, the query is built from the pooled responses of the unmeasured drugs, and the retriever selects the next batch of candidates. In all three, query states are obtained by k -means with $k = 2$ on the query cells and the minority state is the smaller cluster; candidate populations are capped at 120–150 cells per drug.

The response-matching criterion is an energy-based welfare regret. Each candidate’s welfare is the

aggregate over query states s of its negative energy distance to that state, $U(d) = \text{agg}_s(-\mathcal{E}(P_d, Q_s))$ with the aggregator taken as the minimum in the worst-case form, and the regret of the selected candidate $\hat{d}$ is

$$\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0,$$

where $\mathcal{L}$ is the observed library, so the reference is the best candidate actually available rather than an unobservable optimum. Minority-state coverage is reported on the same rankings as the cosine between the candidate’s mean response and the mean response of the query’s minority state, rescaled from $[-1, 1]$ to $[0, 1]$. Regret is zero when the top-ranked candidate is optimal under that objective and increases as the selected candidate becomes less favourable. We report regret reduction relative to mean-signature retrieval, such that a positive value indicates lower regret under this response-matching criterion. This evaluation is intentionally closely aligned with the population-level information used by the retriever and is therefore interpreted as evidence of response fidelity rather than as an independent functional outcome.

Predict-then-rank evaluation. The predictor-generated retrieval reported in Fig. 5b factorizes the pipeline the field actually uses into a forward predictor and a retrieval rule, so that the retrieval layer can be varied with the predictor held fixed. Queries are cross-line: for each of the three SciPlex3 cell-line pairs (K562–A549, A549–MCF7, K562–MCF7) and each of ten drugs selected by a cross-context divergence probe, the query is that drug’s response blended across the two lines at mixing fraction $\alpha \in \{0.5, 0.7, 0.9\}$ over 200 cells, with eight seeds, giving 720 queries. The candidate library is the query’s own drug together with up to 30 further drugs shared by both lines; the query’s drug is the single relevant candidate. Each predictor is asked to imagine every candidate’s response in the query’s real blended control cells, mixed at the same α , so no predictor sees the within-query line structure. The imagined populations are then ranked by four retrieval rules: mean cosine, R^2 of the predicted against the observed mean delta, energy distance and coverage. Performance is nDCG@10 with binary relevance, and the quantity plotted is the distributional minus the mean-cosine value for the same predictor, so each predictor is its own control.

Task-proximal biological evaluation. The same fixed rankings were next evaluated using mechanism-of-action (MoA) recovery and minority-state coverage. MoA recovery was quantified by normalized discounted cumulative gain (nDCG)⁴⁰, which rewards candidates sharing the query perturbation’s annotated mechanism when they occur near the top of the ranked list. The comparison is defined only when the candidate library contains members of the query mechanism class. Of the 765 partial-observation queries, 600 satisfy this requirement. The remaining 165 queries—120 leave-MoA-out queries and 45 partial-library queries—are excluded from MoA analyses rather than assigned an artificial zero, because the relevant mechanism is absent from the candidate library and recovery is therefore undefined.

Minority-state coverage asks whether the top-ranked candidate better preserves the response of the less represented cellular state in the controlled heterogeneous-population construction. As with MoA recovery, this criterion is applied to rankings that have already been generated; the evaluator does not alter the query, the retrieval score or the candidate ordering. Differences between population-level and mean-signature retrieval are therefore attributable to how the same rankings score under the biological

520 criterion rather than to refitting or re-ranking.

521 **Real-data projection.** To examine whether the minority-state result extends beyond the controlled
522 partial-observation setting, we evaluated the top-ranked population-level and mean-signature candi-
523 dates across 239 real-data tasks. These comprised 90 constructed cross-cell-line SciPlex3 tasks, 90 within-
524 cell-line SciPlex3 tasks, 23 Frangieh tasks, 12 CD34+ tasks and 24 tasks using predicted candidate pop-
525 ulations. For each task we report the difference in minority-state coverage between the population-level
526 and mean-signature selections.

527 The 239 tasks are treated as a distribution of task-level differences rather than as independent bio-
528 logical replicates of a single experiment. In particular, the constructed SciPlex3 tasks share data source,
529 preprocessing and candidate structure, and the three SciPlex3 cell lines constitute leave-one-cell-line-out
530 variation within one experiment rather than leave-one-dataset-out generalization. Thresholded counts
531 of “distributionally dominant” tasks are therefore not used as a primary result; the continuous difference
532 in coverage is reported instead.

533 **Controlled synthetic benchmark**

HIR-Bench was used as a controlled model of when within-population structure can change the preferred retrieval candidate. The benchmark generates heterogeneous retrieval tasks from a two-subpopulation latent welfare model with tunable subpopulation balance and conflict between subpopulation-specific preferences. Under this construction, the preferred candidate changes at the analytical boundary

$$\alpha^* = \frac{B}{A+B},$$

534 where the notation follows the latent welfare specification of the generator. The boundary therefore
535 provides a known-ground-truth distinction between regimes in which the population mean is sufficient
536 to preserve candidate preference and regimes in which resolving the two subpopulations can change
537 that preference.

538 Each parameter configuration was instantiated under three information conditions: fully observed
539 response populations, predicted populations containing only mean-level response information, and pre-
540 dicted populations retaining population structure. The full benchmark comprises 672 parameter config-
541 urations evaluated over 20 random seeds, giving 13,440 instances. HIR-Bench is used to establish the
542 existence and observability of a controlled preference-changing regime; it is not treated as a generative
543 model of any specific biological dataset.

544 To ask whether entry into this regime could be recognized from information available at retrieval
545 time, we fitted logistic-regression classifiers to predict whether population structure changes the pre-
546 ferred candidate. The observable feature set contains only statistics that can be computed from the
547 query and candidate response populations: subpopulation-variance ratio, isotropy index and response
548 diversity. For comparison, an oracle-derived feature set contains top- k disagreement and Kendall conflict
549 scores computed from the latent utility matrix used by the generator. These oracle-derived quantities are
550 not available to a retrieval system and are included only as a controlled comparison.

551 Cross-validation was grouped by the parameter pair $(\alpha, \text{conflict})$, because the retrieval-flip label is
552 determined by these two variables. This gives 28 label-determining parameter cells, which are treated

as the effective independent units. Instances generated from the same parameter cell are therefore never divided between training and test sets. Because each held-out parameter cell contains a single outcome class by construction, AUC is computed from the pooled out-of-fold predictions rather than within individual folds.

We additionally examined whether the synthetic regime boundary could be transferred to the real-data tasks. The method-performance layer required for this projection was available only on a reduced grid; the resulting classifier was compared against a constant majority-class baseline on the real-data projection (Fig. 4b). On the basis of that comparison we do not use HIR-Bench to assign real biological datasets to a population-favourable or mean-sufficient regime. In the main text, it serves only as a controlled demonstration that population structure can, in principle, become consequential for candidate choice.

External functional evaluation using GDSC2

To test whether transcriptomic retrieval rankings capture information that extends beyond the single-cell expression measurements used to construct them, we evaluated the SciPlex3 rankings against drug-response profiles from the Genomics of Drug Sensitivity in Cancer (GDSC2) resource^{30,31}. The external criterion was defined as similarity between the functional response profiles of two drugs rather than as absolute drug potency, because the retrieval task asks which candidate produces a response most similar to the query rather than which candidate is most cytotoxic.

SciPlex3 compounds were matched to GDSC2 by normalized drug name, with substring matching used only as a fallback. After removal of one duplicate mapping and restriction to the GDSC2 screening platform, 35 compounds remained. GDSC1 and GDSC2 measurements were not pooled because they derive from different assay and dose-range settings. For each matched compound, its functional profile was the vector of fitted dose-response AUC values across GDSC2 cell lines. The three cell lines used for the transcriptomic retrieval experiments—A549, K562 and MCF7—were excluded from these profiles, so the functional evaluation was performed in biological contexts not used to construct the RNA rankings.

Cell lines differ systematically in their overall sensitivity to drug treatment, which can induce positive correlations between otherwise unrelated compound profiles. We therefore removed this shared cell-line effect before calculating drug-drug similarity. For cell line l and compound d , the centred response was

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in cell line l , rather than only the 35 compounds overlapping SciPlex3. Functional similarity between query drug q and candidate drug c was then defined as

$$F(q, c) = \rho_s \left(\widetilde{\text{AUC}}_{q,\bullet}, \widetilde{\text{AUC}}_{c,\bullet} \right),$$

where ρ_s denotes Spearman correlation across cell lines shared by the two compounds. Drug pairs measured in fewer than 100 common cell lines were excluded. This yielded 591 usable drug pairs, with a median of 909 shared cell lines per pair. The corresponding uncentred functional similarity was retained as a sensitivity analysis.

For each SciPlex3 cell line, retrieval was evaluated in a leave-one-drug-out design. The 10 μM re-

583 sponse population of the held-out compound served as the query, and the remaining matched com-
 584 pounds formed the candidate library, giving 34–35 candidates per query. Each RNA-based retrieval rule
 585 produced a candidate ranking without access to GDSC2 data. Agreement between that ranking and the
 586 external functional ordering $F(q, c)$ was quantified by Spearman correlation across candidates.

The primary RNA-derived rankings were energy-distance population retrieval and control-referenced mean-cosine retrieval. To test whether apparent functional agreement required cell-to-cell distributional information, we additionally used a response-magnitude matching score,

$$M(q, c) = - \left| \|\bar{d}_q\| - \|\bar{d}_c\| \right|,$$

587 which prefers candidates whose overall transcriptional response magnitude is similar to that of the query
 588 but uses no within-population structure. A query-independent magnitude score $\|\bar{d}_c\|$ was retained as a
 589 diagnostic control. An uncorrected mean-cosine score based on treated means without matched-control
 590 subtraction and a potency-matching diagnostic were analysed as additional sensitivity checks rather
 591 than as primary retrieval baselines.

592 To estimate the component of the energy-based association not explained by response magnitude, we
 593 computed a partial rank correlation with the magnitude-matching channel removed. Retrieval scores
 594 and functional similarities were rank-transformed, each was residualized with respect to $M(q, c)$, and
 595 the correlation between the two residuals was reported as the magnitude-adjusted association.

596 Per-query Spearman correlations are summarized by their median within each SciPlex3 cell line.
 597 Queries from the same cell line share the same candidate library and are therefore not treated as in-
 598 dependent biological replicates. The three cell lines provide the higher-level analysis unit; query-level
 599 paired tests are reported only as descriptive analyses where indicated.

600 **Protein-based evaluator-shape experiment**

601 Frangieh Perturb-CITE-seq³² provides matched RNA and surface-protein measurements from the same
 602 cells, allowing the statistical form of an external evaluator to be changed while holding its biological
 603 measurement fixed. RNA and protein matrices were joined by cell barcode before analysis; row position
 604 was not used because the released modality-specific matrices differ in cell number and ordering. Of the
 605 24 measured antibodies, four isotype controls were removed, leaving 20 surface-protein features. Protein
 606 counts were centred-log-ratio normalized across antibodies within each cell.

607 All analyses were performed separately within the Control, IFN γ and Co-culture conditions to avoid
 608 mixing immune context with perturbation identity. Knockouts represented by at least 60 barcode-
 609 matched cells were retained, and populations were subsampled to at most 250 cells using a fixed random
 610 seed.

611 **Protein-based evaluators.** For knockout k in immune condition c , let $\bar{d}_{k,c}^{\text{prot}}$ denote its mean control-
 612 referenced protein response and let $D_{k,c}^{\text{prot}}$ denote the corresponding set of per-cell protein responses. The
 613 mean-based evaluator was defined as

$$E_{\text{mean}}(q, d) = \cos \left(\bar{d}_{q,c}^{\text{prot}}, \bar{d}_{d,c}^{\text{prot}} \right).$$

614 The population-based evaluator was defined from the same cells and protein measurements as

$$E_{\text{pop}}(q, d) = -\mathcal{E} \left(D_{q,c}^{\text{prot}}, D_{d,c}^{\text{prot}} \right),$$

where $\mathcal{E}$ is the energy distance. The negative sign converts distance to similarity so that larger values indicate a closer match under both evaluators.

For both definitions, protein responses were referenced to the matched control population from the same immune condition. Thus,

$$\bar{d}_{k,c}^{\text{prot}} = \text{mean} \left(X_{k,c}^{\text{prot}} \right) - \mu_{0,c}^{\text{prot}},$$

and

$$D_{k,c}^{\text{prot}} = \left\{ x_{i,k,c}^{\text{prot}} - \mu_{0,c}^{\text{prot}} \right\}_{i=1}^{n_{k,c}}.$$

Using the same matched control for the two evaluator forms ensures that their difference arises from whether protein responses are summarized by their mean or retained as a cellular population, rather than from a change in biological reference.

Fixed RNA-derived rankings. The RNA-derived rankings were computed once and then evaluated against both protein criteria. The primary rankings were energy-distance population retrieval and control-referenced mean-cosine retrieval. The query populations, candidate libraries and RNA-derived candidate scores were identical under the two protein evaluators, and no protein measurement was available to any retrieval method. The experiment therefore changes only the statistical representation used to judge the same RNA-derived ranking.

Response-magnitude control. Because a population distance can also encode differences in overall response magnitude, we included a query-dependent scalar control that uses no cell-to-cell distributional information. For query q and candidate d , magnitude matching was defined as

$$M(q, d) = - \left| \left| \bar{d}_{q,c}^{\text{RNA}} \right| - \left| \bar{d}_{d,c}^{\text{RNA}} \right| \right|.$$

This ranking favours candidates with a transcriptional response magnitude similar to that of the query while discarding all within-population structure. The same magnitude ranking was evaluated against both protein criteria.

For each query, agreement between an RNA-derived ranking and a protein-based evaluator was quantified by Spearman correlation across candidate knockouts. Query-level correlations were summarized by their median within each immune condition. The three immune conditions were reported separately and treated as the higher-level analysis unit; queries within a condition share candidate structure and were not interpreted as independent biological replicates.

Differential-response analysis of predicted populations

We next asked whether perturbation predictors preserve the state-specific response variation that population-level retrieval is intended to exploit. These analyses were not designed as a benchmark of

643 perturbation-prediction models. Instead, the predictors were used as alternative sources of candidate
644 response populations to test which forms of response structure remain available after prediction.

645 **Predict-then-rank response sources.** Three predictor families were used in the predict-then-rank anal-
646 ysis. The average-effect predictor estimates a mean perturbation effect for each drug and applies that
647 effect to cells in the query context. The nearest-neighbour predictor transfers the perturbation effect
648 from the training context whose untreated control state is most similar to the query context. The linear-
649 latent predictor implements latent arithmetic in which a perturbation shift is applied in a learned latent
650 representation before decoding cells back to expression space. These models were used only to gener-
651 ate candidate populations for retrieval and were not interpreted as representative benchmarks of the
652 broader virtual-cell model class.

653 To construct a predicted population, the predicted perturbation effect was applied to the real control
654 cells of the query context one cell at a time. Thus, a predicted population retained the cell-to-cell variation
655 already present in the untreated population whenever the predictor permitted that variation to propa-
656 gate through the model. We did not use synthetic Gaussian clouds of the form control mean $+ \delta + \epsilon$ for
657 differential-response analyses, because such a construction imposes its own population geometry and
658 can obscure whether the predictor itself generates state-dependent effects.

659 The reported predict-then-rank experiment was fitted using the full SciPlex3 matrix, and the cell lines
660 contributing to the query populations were not excluded from predictor fitting. These results should
661 therefore be interpreted as an in-sample ceiling on the candidate-generation step rather than as evidence
662 of cross-context predictive generalization.

663 **Within-context definition of differential response.** Differential response was measured within a sin-
664 gle cellular context so that variation between contexts could not be mistaken for state-specific pertur-
665 bation effects. For each SciPlex3 context, up to 400 untreated control cells were represented by 30-
666 component PCA and partitioned into two cellular states using k -means with $k = 2$. Subpopulation
667 sizes and silhouette scores were retained as diagnostics of the underlying state partition.

668 For perturbation d , each predictor was applied separately to the control cells belonging to the two
669 states. The induced response of state k was defined as

$$r_{d,k} = \text{mean} \left(\hat{X}_{d,k} \right) - \text{mean} \left(X_{0,k} \right),$$

670 where $X_{0,k}$ denotes the untreated cells assigned to state k and $\hat{X}_{d,k}$ denotes their predicted post-
671 perturbation profiles. Differential response was summarized by the cosine between the two state-specific
672 response vectors,

$$R_d = \cos \left(r_{d,0}, r_{d,1} \right).$$

673 A value near 1 indicates that the two cellular states are predicted to respond in nearly the same
674 direction, whereas lower values indicate increasingly state-specific perturbation responses. Real treated
675 cells from the same context, partitioned according to the same control-derived state definition, were
676 analysed identically and provide the observed reference.

677 **Additive-response limit.** The limiting behaviour of an additive predictor follows directly from its
 678 model form. If the same perturbation effect δ_d is applied to every cell,

$$\hat{x}_{i,d} = x_i + \delta_d,$$

679 then the induced response is identical in every cellular state,

$$r_{d,0} = r_{d,1} = \delta_d,$$

680 and therefore

$$\cos(r_{d,0}, r_{d,1}) = 1.$$

681 An additive predictor may therefore preserve substantial heterogeneity in the untreated population
 682 while generating no state-specific variation in the perturbation effect itself. This distinction is why pop-
 683 ulation heterogeneity and differential perturbation response are measured separately.

684 **Latent-model reconstruction reference.** For predictors that reconstruct the input before applying a
 685 latent perturbation shift, the measured response depends on the reference used to define the perturbation
 686 effect. Let $P(C_k)$ denote the model reconstruction of control state C_k , and let $W\delta_z$ denote the decoded
 687 latent perturbation shift. Referencing the predicted population to the observed control cells gives

$$r_k^{\text{obs}} = \text{mean}(P(C_k)) - \text{mean}(C_k) + W\delta_z,$$

688 whereas referencing it to the model's own reconstructed control gives

$$r_k^{\text{model}} = W\delta_z.$$

689 The first quantity includes state-dependent reconstruction error; the second isolates the perturba-
 690 tion effect represented by the model. Both references were therefore retained for latent predictors. The
 691 additive-limit identity above applies to the model-referenced response, for which a shared latent shift
 692 necessarily yields a cosine of 1.

693 **Additional predictor diagnostics.** The within-context differential-response analysis also included a
 694 deployed scGen model and a non-parametric optimal-transport mapping to test whether predictors not
 695 restricted to the simple gene-space average-effect construction generated response variation closer to
 696 that observed in real treated populations. These models were evaluated with the same state partition
 697 and induced-response definition described above.

698 The scGen model is the published implementation⁶ (scgen.SCGEN on scvi-tools⁴¹), trained on the
 699 SciPlex3 tensor with the perturbation label as the batch key and the cell line as the label key, 30 latent
 700 dimensions, batch size 512, at most 40 epochs with early stopping after 15 epochs without improvement,
 701 and a fixed seed. Its own latent-arithmetic rule is used verbatim: one latent delta per perturbation,
 702 estimated as the mean latent of treated cells minus the mean latent of control cells, and a prediction
 703 is formed by encoding the query context's control mean, adding that delta and decoding. Because the

704 decoder is nonlinear, a single latent delta decodes to a different gene-space response for each cell, so the
705 induced-response cosine is not pinned to 1 by construction as it is for the linear-latent control.

706 The optimal-transport map is an entropic (Sinkhorn) coupling³³ solved with POT⁴² between control
707 and treated cells, with uniform marginals, regularization 0.05, at most 500 iterations, and a cost matrix
708 rescaled by its maximum. It is solved in a 30-component PCA latent fitted unsupervised on at most
709 20,000 cells, because in the full 2,000-gene space all pairwise costs are nearly equal after rescaling and
710 the plan carries no signal; the gene-space variant did not clear the learning check and was not used. Each
711 fitting context is centred on its own control mean before the coupling is solved, so the map transports
712 response rather than baseline identity. The map is fitted only on contexts other than the query’s, so the
713 query context’s treated cells are never seen. A barycentric projection of the coupling gives a per-source-
714 cell displacement, which is extended to the query’s real control cells by nearest-source matching in the
715 same centred latent space and decoded through the PCA inverse transform.

716 Both models are held to a pre-specified learning check: a predictor whose transferred mean-delta
717 cosine falls below 0.30 is declared void and is not interpreted. scGen (0.524) and the optimal-transport
718 map (0.352) pass this criterion. Each predictor’s induced response is measured against its own zero-effect
719 output rather than against real control cells, which cancels the reconstruction bias that latent models
720 otherwise contribute to this quantity. Values are means over 30 cell-line and drug combinations (three
721 cell lines, ten drugs, 400 control cells per context split into two states).

722 No model evaluated here was used to establish a general ranking of perturbation-prediction architec-
723 tures. The analysis tests a narrower requirement: whether a predicted population contains state-specific
724 perturbation effects that could, in principle, be exploited by a downstream population-aware retriever.

725 Recoverability of state-specific responses

726 Differential responses can contribute to population-level retrieval only if the corresponding cellular
727 structure can be recovered from the observed response population. We therefore measured recover-
728 ability by comparing unsupervised partitioning with a supervised estimate of how well the same states
729 could be distinguished when their identities were known. This comparison separates failure of an unsu-
730 pervised method from absence of discriminative information in the measured representation.

731 **Controlled two-state response mixtures.** The primary recoverability experiment used K562 cells
732 treated at 10 μ M with compounds from HDAC and JAK perturbation classes. Response populations
733 from the two sources were combined to form mixtures with known two-state composition. Between
734 50 and 400 cells were sampled from each source over 10 random seeds. Ground-truth source identity
735 was used only for evaluation and for the supervised ceiling; it was not available to the unsupervised
736 procedures.

737 All methods were evaluated using best-permutation accuracy so that supervised and unsupervised
738 procedures shared the same outcome scale. For a clustering assignment z_i and source label y_i , cluster
739 identities were matched to source identities after fitting, and accuracy was computed as

$$A = \max_{\pi} \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i = \pi(z_i)],$$

740 where π denotes the permutation of cluster labels that maximizes agreement with the known sources.

For methods returning more than two clusters, each cluster was assigned to its majority source label. HDBSCAN noise points were treated as an additional cluster before label matching. This convention favours rather than penalizes over-partitioning and therefore provides a conservative comparison with the supervised ceiling.

Unsupervised recovery. Four unsupervised procedures were evaluated: k -means, Gaussian-mixture modelling, Leiden clustering⁴³ and HDBSCAN⁴⁴. The first two methods were given the known number of source populations, $k = 2$, whereas Leiden and HDBSCAN determined their own partition structure. Unless otherwise stated, clustering was performed in a 30-component PCA representation fitted to the complete response mixture. The best unsupervised accuracy across the evaluated methods was retained as the recoverable structure extracted without source labels.

Supervised ceiling. To estimate how much source information was present in the same expression representation, we trained logistic regression, random forest and k -nearest-neighbour classifiers using the known source labels. Models were evaluated by five-fold stratified cross-validation, and the supervised ceiling for each mixture was defined as the best out-of-fold accuracy among the three classifiers.

Because the clustering procedures operate on a PCA representation fitted to the complete mixture, we report two supervised references. In the leak-free reference, PCA was re-fitted within each training fold and test cells were projected without contributing to the fitted representation. In the matched reference, the classifiers received the same transductive PCA representation as the unsupervised methods; PCA itself does not use source labels. The matched representation provides the primary like-for-like comparison between supervised and unsupervised recovery, whereas the in-fold representation is retained as a sensitivity analysis.

Recoverability was summarized by the paired difference

$$G = A_{\text{sup}} - A_{\text{unsup}},$$

where A_{sup} is the supervised ceiling and A_{unsup} is the best unsupervised accuracy for the same mixture. A small value of G indicates that little discriminative information remains beyond what unsupervised partitioning already extracts. A larger value indicates that additional response information is present in the representation but is not recovered by the unsupervised procedure. The gap is therefore interpreted jointly with the absolute supervised ceiling rather than as a stand-alone measure.

Drug-pair resolution. The pooled HDAC-versus-JAK construction combines cells from several compounds within each class and therefore introduces within-class variation unrelated to the distinction between two individual drug responses. We consequently repeated the analysis at drug-pair resolution. Individual HDAC-versus-JAK drug pairs were analysed separately in K562 at 10 μ M using the same representation, unsupervised methods, supervised probes and best-permutation accuracy. Twenty-four drug pairs were sampled with a fixed seed from the 513 available pairs. This analysis tests whether an apparently low recoverability ceiling reflects genuinely weak response separation or the additional heterogeneity introduced by pooling several drugs into broad mechanism classes.

776 **Sensitivity to state separation.** To verify that the recoverability analysis can detect additional infor-
777 mation when it is present, we constructed a positive-control series in which separation between the two
778 source populations was increased by a factor

$$s \in \{1, 1.5, 2, 3, 5, 8\}.$$

779 The transformation was defined once from the full source populations and then applied before cell
780 subsampling, so the separation manipulation was not fitted to the particular cells used for evaluation.
781 The unmodified data correspond to $s = 1$. Increasing s provides a controlled test of whether the su-
782 pervised ceiling separates from unsupervised recovery as the two response states become increasingly
783 distinguishable.

784 The same supervised-versus-unsupervised framework was subsequently applied to naturally het-
785 erogeneous glioblastoma responses and to Tahoe-100M, with dataset-specific sampling and partition
786 definitions described below. The recoverability analysis therefore measures an information requirement
787 rather than a particular clustering algorithm: whether state-specific response information is present in
788 the population, and how much of that information remains unextracted without labels.

789 **Natural heterogeneity and large-scale validation**

790 We next examined the information requirements for population-level retrieval in datasets in which cellu-
791 lar heterogeneity was not created by mixing experimentally distinct source populations. Patient-derived
792 glioblastoma was used to test the requirements in naturally heterogeneous tissue, and Tahoe-100M was
793 used to assess the same quantities at substantially larger scale. Neither dataset was used as an additional
794 retrieval benchmark.

795 **Patient-derived glioblastoma.** The ZhaoSims2021 dataset contains acute-slice cultures and surgical
796 biopsies from ten glioblastoma patients exposed to six compounds: Ana-12, Ipsenisib, RO4929097,
797 Tazemetostat, etoposide and panobinostat. Expression counts were normalized to 10^4 per cell, log1p-
798 transformed and reduced to 2,000 highly variable genes. Because only one patient received more than
799 two compounds, the cohort does not support a sufficiently replicated leave-one-drug-out retrieval
800 benchmark. We therefore use it only to measure state-specific response, recoverability and the extent to
801 which population-level similarity differs from compartment-specific response ordering.

802 **Cellular-compartment assignment.** The released dataset does not provide resolved cell-type annota-
803 tions for the analysed cells. Compartments were therefore assigned from canonical marker sets. Marker
804 scores were computed from the full log-normalized expression matrix before highly variable gene selec-
805 tion. A cell was assigned to a compartment only when that compartment had the highest marker score,
806 the mean expression of its defining markers was at least 0.25, and its marker-score z-score exceeded the
807 second-highest compartment by at least 0.25. Cells failing either criterion were labelled unassigned and
808 excluded.

809 This procedure retained three compartments that passed marker-expression validation: malignant
810 glioma cells, tumour-associated myeloid cells and oligodendrocytes. Overall, 39.9% of cells were ex-
811 cluded as unassigned. Sensitivity analyses varied both assignment thresholds between 0.10 and 0.50

812 and gave similar differential-response and recoverability estimates; these analyses are reported in the
 813 Supplementary Information. Compartment labels were used only to define and evaluate the biological
 814 states and were never supplied to an unsupervised recovery procedure.

815 **Differential response in natural tissue.** State-specific perturbation responses were measured between
 816 malignant and myeloid compartments. For patient p , drug d and compartment k , the response vector
 817 was

$$r_{p,d,k} = \text{mean} \left(X_{p,d,k}^{\text{treated}} \right) - \text{mean} \left(X_{p,k}^{\text{control}} \right).$$

818 Only patient–drug combinations with at least 50 treated cells in both compartments were included.
 819 Controls were matched within both patient and compartment. Differential response was summarized as

$$R_{p,d} = \cos \left(r_{p,d,\text{malignant}}, r_{p,d,\text{myeloid}} \right).$$

820 This construction removes pre-existing differences between patients and cellular compartments be-
 821 fore comparing the directions of their drug-induced responses.

822 **Recoverability in natural tissue.** Recoverability was evaluated as a drug-response discrimination
 823 problem rather than as a cell-type classification problem. For each eligible patient and compartment,
 824 cells treated with drug a were distinguished from cells treated with drug b . Each drug contributed
 825 at least 50 cells and was capped at 150 cells per split; five random seeds were used. This yielded 36
 826 within-patient drug-pair splits across four patients and two cellular compartments.

827 The same protocol described above for controlled mixtures was then applied: unsupervised meth-
 828 ods operated in a common 30-component PCA representation, source recovery was evaluated by best-
 829 permutation accuracy, and the supervised ceiling was defined as the best five-fold cross-validated ac-
 830 curacy among logistic regression, random forest and k -nearest neighbours. The primary recoverability
 831 gap was computed using the matched PCA representation shared by the supervised and unsupervised
 832 procedures,

$$G_j = A_{\text{sup}}^{(j)} - A_{\text{unsup}}^{(j)},$$

833 for drug-pair split j . The reported cohort-level gap is the median of the paired values $\{G_j\}$ rather
 834 than the difference between the marginal supervised and unsupervised medians.

835 Because the 36 splits are not statistically independent, uncertainty was estimated at the patient level
 836 rather than by treating drug pairs as independent replicates. Cluster bootstrap resampling was per-
 837 formed over patients, and the analysis was additionally repeated after excluding the patient contribut-
 838 ing the largest number of drug-pair comparisons. Fixed supervised–unsupervised method pairs were
 839 also examined as a sensitivity analysis to the use of the best-performing method within each family.

840 **Proxy for downstream decision relevance.** The glioblastoma cohort does not contain enough drug
 841 coverage to test whether population-level retrieval directly changes the preferred candidate. We there-
 842 fore quantified a more limited property: whether drug similarity at the whole-population level preserves

843 the ordering observed within a specific cellular compartment.

844 For each pair of drugs measured in the same patient, we first computed the cosine similarity be-
845 tween their population-mean response signatures and the cosine similarity between their malignant-
846 compartment response vectors. Agreement between these two similarity structures was summarized by
847 Spearman correlation across drug pairs.

848 Because the whole-population mean contains malignant cells and therefore shares observations with
849 the malignant-compartment response, we additionally used a disjoint comparison. For drug pair (a, b) ,
850 myeloid-response similarity was defined as

$$S_{a,b}^{\text{myeloid}} = \cos(r_{a,\text{myeloid}}, r_{b,\text{myeloid}}),$$

851 and malignant-response similarity as

$$S_{a,b}^{\text{malignant}} = \cos(r_{a,\text{malignant}}, r_{b,\text{malignant}}).$$

852 We then computed

$$\rho_{\text{disjoint}} = \rho_S(S_{a,b}^{\text{myeloid}}, S_{a,b}^{\text{malignant}}),$$

853 using only drug pairs available within the same patient. The two sides of this comparison contain no
854 shared cells and each compartment is referenced to its own matched control. This statistic is interpreted
855 as a proxy for the amount of candidate-ordering headroom available to population structure, not as a
856 direct measure of retrieval performance. No pass-fail threshold was applied.

857 **Large-scale validation in Tahoe-100M.** Tahoe-100M¹⁶ was used to re-measure the same information
858 requirements under unconstructed within-cell-line heterogeneity at larger scale. We analysed plate 3,
859 which contains a complete grid of 50 cell lines and 93 compounds at one concentration per compound,
860 together with a DMSO_TF vehicle condition in every cell line. The released plate contains 4,158,278 cells
861 with a median of 648 cells per cell-line-compound condition. We used the released 2,304-gene log1p-
862 transformed representation without additional feature selection or representation fitting.

863 Two independently defined cellular-state partitions were evaluated. The first used the released cell-
864 cycle annotation, comparing G1 with G2M cells and excluding S-phase cells as an intermediate state.
865 The second was defined by fitting k -means with $k = 2$ to the control cells of each cell line and assigning
866 treated cells to the nearest control-derived centroid. The latter partition is therefore fixed from the un-
867 treated state before any drug-response measurement is examined. Both partitions were analysed without
868 selecting between them according to downstream performance.

869 For each eligible drug-cell-line condition, the response of state k was defined relative to its own
870 state-matched controls,

$$r_{d,c,k} = \text{mean}(X_{d,c,k}^{\text{treated}}) - \text{mean}(X_{c,k}^{\text{control}}).$$

871 Differential response was measured as the cosine between the two state-specific response vectors,

$$R_{d,c} = \cos(r_{d,c,0}, r_{d,c,1}).$$

872 A condition was included only when at least 50 cells were available in each required treated and
873 control state.

874 Tahoe recoverability was evaluated using drug identity rather than the state labels used above.
875 Within each cell line, 20 random pairs of compounds were sampled, with balanced samples of up to
876 300 cells per compound. Pairs were required to contain at least 100 cells for each compound. The su-
877 pervised reference was five-fold cross-validated logistic regression. The unsupervised arm included
878 k -means in gene space, k -means and Gaussian mixtures in 50-dimensional PCA space, and Leiden clus-
879 tering with its resolution reduced until two communities were returned. All methods were evaluated by
880 best-permutation accuracy, and the recoverability gap was defined as above.

881 Finally, a disjoint state-ordering statistic was used as a large-scale proxy for downstream decision
882 relevance. Within each cell line, cosine similarity was computed for every eligible pair of drug responses
883 separately in the two non-overlapping cellular states. Spearman correlation between the two resulting
884 drug–drug similarity matrices was then calculated,

$$\rho_c = \rho_s(S_{a,b}^{(0)}, S_{a,b}^{(1)}).$$

885 High correlation indicates that the two cellular states preserve similar relative ordering among drug
886 responses, whereas lower correlation leaves greater scope for state-specific information to change candi-
887 date preference. As in the glioblastoma analysis, this quantity does not directly test whether population-
888 level retrieval improves a drug ranking.

889 Tahoe-100M was therefore used only to measure differential response, recoverability and this
890 decision-relevance proxy at larger scale. No retrieval algorithm was run on Tahoe-100M, and no Hit@ k ,
891 regret or external functional retrieval metric was derived from this dataset.

892 Statistical analysis

893 Statistical units were defined by the experimental design rather than by the number of cells or query-
894 candidate comparisons. Cells contribute to the estimation of a perturbation-response population but are
895 not treated as independent biological replicates of retrieval performance⁴⁵. Likewise, repeated queries
896 sharing the same candidate library or biological context are interpreted at the level of that shared exper-
897 imental unit where appropriate.

898 Paired comparisons between retrieval methods evaluated on the same queries were performed us-
899 ing two-sided Wilcoxon signed-rank tests. Comparisons between independent groups used Mann-
900 Whitney U tests, and monotonic associations were quantified using Spearman rank correlation. Where
901 the same hypothesis was examined across multiple pre-specified strata, P values were adjusted using
902 the Benjamini–Hochberg false-discovery-rate procedure⁴⁶. Effect sizes are reported together with signif-
903 icance tests, and statistical significance is not interpreted as evidence of practical importance when the
904 corresponding effect is small relative to the operating range of the metric.

905 For the partial-observation analyses, comparisons between population-level and mean-signature re-
906 trieval were paired within query because the two methods were evaluated on the same query and can-
907 didate library. Mechanism-of-action analyses were restricted to the 600 queries for which the relevant

mechanism class remained represented in the candidate library; undefined queries were excluded rather than encoded as zero. Stratified analyses of response divergence used Benjamini–Hochberg correction across strata. Where observed effects were small, achieved power and the sample size required to reach 80% power were reported to distinguish weak evidence from a precisely estimated negligible effect.

For the GDSC2 functional analysis, Spearman correlations were first computed separately for each query across its candidate drugs and then summarized within each SciPlex3 cell line. Queries from the same cell line share a candidate library and are therefore not independent biological replicates. Query-level paired tests are reported only where indicated and are explicitly treated as nominal; the three SciPlex3 cell lines provide the higher-level analysis unit for the functional comparison. The protein-based evaluator-shape experiment was analysed analogously: per-query rank correlations were summarized within each of the Control, IFN γ and Co-culture conditions, and the three immune conditions were retained as the higher-level units rather than pooling all knockout queries as independent replicates.

For HIR-Bench, repeated random seeds generated from the same parameter configuration were not treated as independent observations for prediction of the retrieval-flip regime. Cross-validation was grouped by the label-determining (α , conflict) parameter pair, giving 28 effective parameter cells. Because each held-out parameter cell contains a single outcome class by construction, classifier performance was summarized by pooled out-of-fold AUC rather than by averaging fold-specific AUC values.

For the patient-derived glioblastoma analysis, drug-pair splits from the same patient were treated as dependent. Uncertainty in the recoverability gap was therefore estimated by cluster bootstrap resampling at the patient level rather than at the drug-pair level, using 5,000 bootstrap resamples. Because only four patients contributed evaluable drug-pair splits, these intervals were interpreted cautiously and were supplemented by repeating the analysis after excluding the patient contributing the largest number of comparisons. Recoverability differences were calculated within each drug-pair split before aggregation, so the reported median gap is the median of paired supervised–unsupervised differences rather than the difference between their marginal medians.

Tahoe-100M analyses were summarized across eligible drug–cell-line conditions or within-cell-line drug pairs, according to the statistic being measured. Cell-level observations were used to estimate state-specific responses and recoverability but were not treated as independent replicates. The two cellular-state definitions—released cell-cycle state and the independently control-derived k -means partition—were analysed in parallel and were not selected according to which produced the more favourable result.

All tests are interpreted in conjunction with the corresponding effect size, sample structure and replication level. No inference from an external functional association, a state-recoverability gap or a decision-relevance proxy is interpreted as evidence of therapeutic efficacy.

Data availability

All datasets analysed in this study are publicly available.

SciPlex3, Frangieh Perturb-CITE-seq and ZhaoSims2021 were obtained in harmonized form from scPerturb¹⁴, Zenodo record 13350497 ([doi:10.5281/zenodo.13350497](https://doi.org/10.5281/zenodo.13350497), release 1.4). The exact harmonized files used in the analyses were SrivatsanTrapnell2020_sciplex3.h5ad for SciPlex3, FrangiehIzar2021_RNA.h5ad and FrangiehIzar2021_protein.h5ad for Frangieh Perturb-CITE-seq, and ZhaoSims2021.h5ad for the patient-derived glioblastoma analysis. The primary SciPlex3 data are avail-

948 able from the Gene Expression Omnibus under accession GSE139944¹⁵; the original Frangieh and
949 ZhaoSims2021 datasets are described in the corresponding publications^{17,32}. The CD34+ haematopoietic
950 progenitor dataset is available from the Gene Expression Omnibus under accession GSE306429.

951 Tahoe-100M¹⁶ was released by Vevo Therapeutics and the Arc Institute under CC0 and is distributed
952 through the Arc Virtual Cell Atlas and Hugging Face (tahoebio/Tahoe-100M). This study uses plate 3
953 only, together with the released obs_metadata and sample_metadata tables. Analyses use the released
954 preprocessed expression matrix comprising 4,158,278 cells and 2,304 highly variable genes in log1p-
955 transformed space. No reported result is derived from the remaining thirteen plates.

956 The external functional evaluation used GDSC2 fitted dose-response data from the Genomics of
957 Drug Sensitivity in Cancer, release 8.5 (GDSC2_fitted_dose_response_270ct23.xlsx), available from
958 <https://www.cancerrxgene.org>. The compound-matching table used to align the SciPlex3 and GDSC2
959 drug panels (drug_match_table.csv) is provided with the analysis code.

960 Processed analysis inputs and the scripts used to construct them from the source datasets are pro-
961 vided in the accompanying repository. Source-data tables underlying each figure panel are provided
962 with the corresponding analysis outputs.

963 Code availability

964 All code used for data processing, retrieval, evaluation and statistical analysis is available at <https://github.com/Boom5426/PopRetrieve>. The repository includes scripts for reconstructing the processed
965 analysis inputs, running the retrieval and evaluation experiments, and regenerating the source-data
966 tables used for the figures. A numerical consistency check re-derives the reported summary statis-
967 tics from the persisted experiment outputs, and a pipeline driver re-executes the main analyses from
968 the processed inputs. The Tahoe-100M analyses are implemented as a self-contained workflow under
969 analysis/tahoe_pilot/. Separate scripts quantify state-specific differential responses, recoverability of
970 drug-response structure and the disjoint state-ordering proxy under both cellular-state definitions, with
971 the corresponding per-condition and per-pair output tables retained for reproducibility.

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975 Author contributions

976 **[[Author contributions, CRediT taxonomy: conceptualization, methodology, software, formal analy-
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978 supervision, funding acquisition.]]**

979 Competing interests

980 **[[Competing interests statement. If none: "The authors declare no competing interests."]]**

981 Additional information

982 Supplementary Information is available for this paper: Supplementary Notes 1–4 and Supplementary
983 Tables 1–4. There are no Extended Data figures: all 21 of their panels are now in Figs. 1–5, and five that

were carried in full by a main panel or a Supplementary Table were removed.
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